ECE4804: VLSI Theory to Tapeout

Systolic Arrays Overview

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Special Acknowledgement

A large portion of the presentation is reproduced from material made available by Dr. Tushar Krishna (GT ECE – Synergy Lab)

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Motivation

- Traditional (Von-Neumann) computing has reached a memory bottleneck
  - Memory transfers take over computation time (e.g. Matrix Multiplication)
- Opportunities in parallelism and memory reuse for Machine Learning (ML) workloads – Matrix Multiplication (MatMul)
  - MatMuls have well-structured memory access patterns
  - Custom hardware (a.k.a. accelerators) can leverage this structure
- But why spend so much effort on ASICs?
  - Opportunity for higher energy efficiency (FLOPS/W)

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Matrix Multiplication (MatMul)

The following slides are courtesy of Dr. Tushar Krishna (GT ECE – Synergy Lab)

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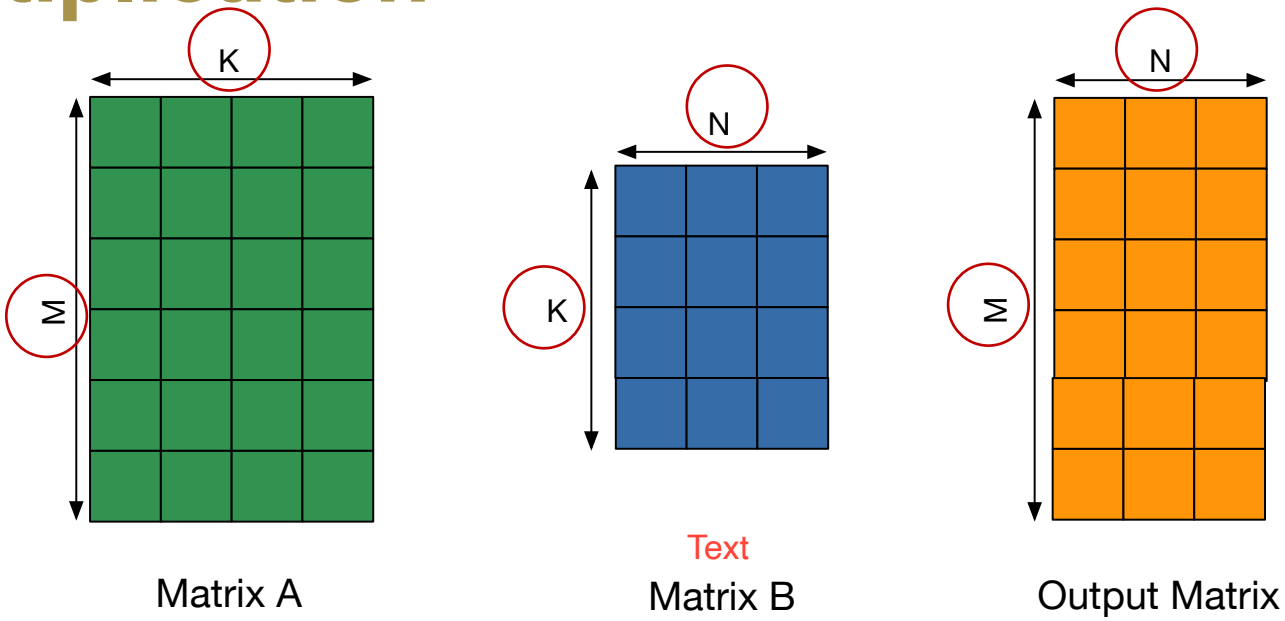
Matrix Multiplication

K: contracted dimension
or contracted rank




```
for(i=0; i<M; i++)
  for(j=0; j<N; j++)
    for(k=0; k<K; k++)
      Output[i][j] += MatrixA[i][k] × MatrixB[k][j];
```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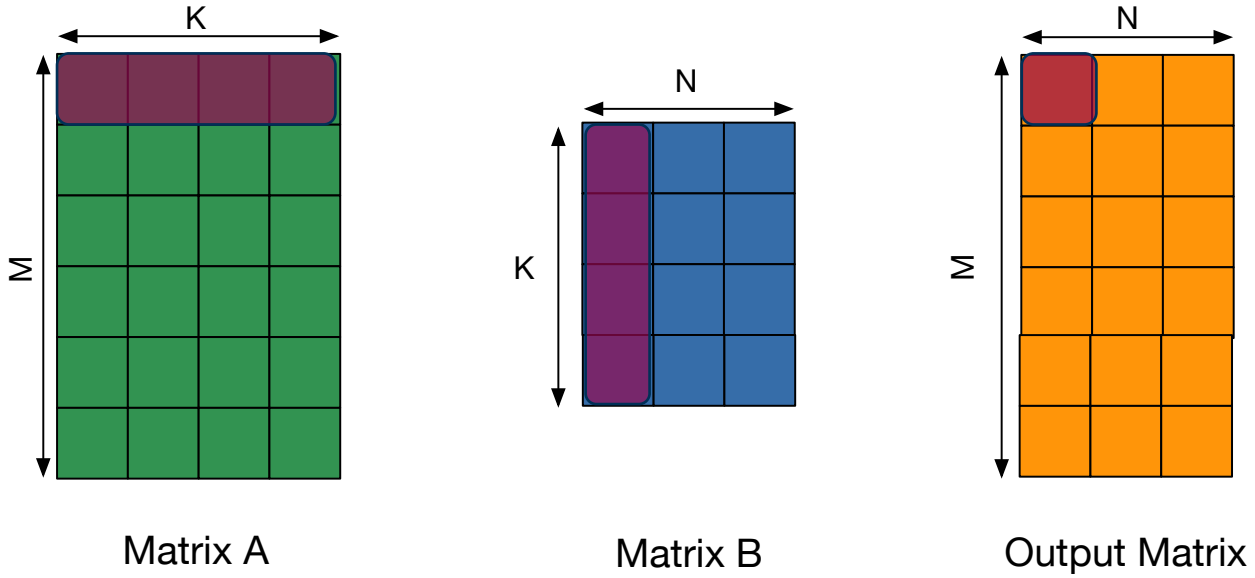
Matrix Multiplication

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for(i=0; i<6; i++)
  for(j=0; j<3; j++)
    for(k=0; k<4; k++)
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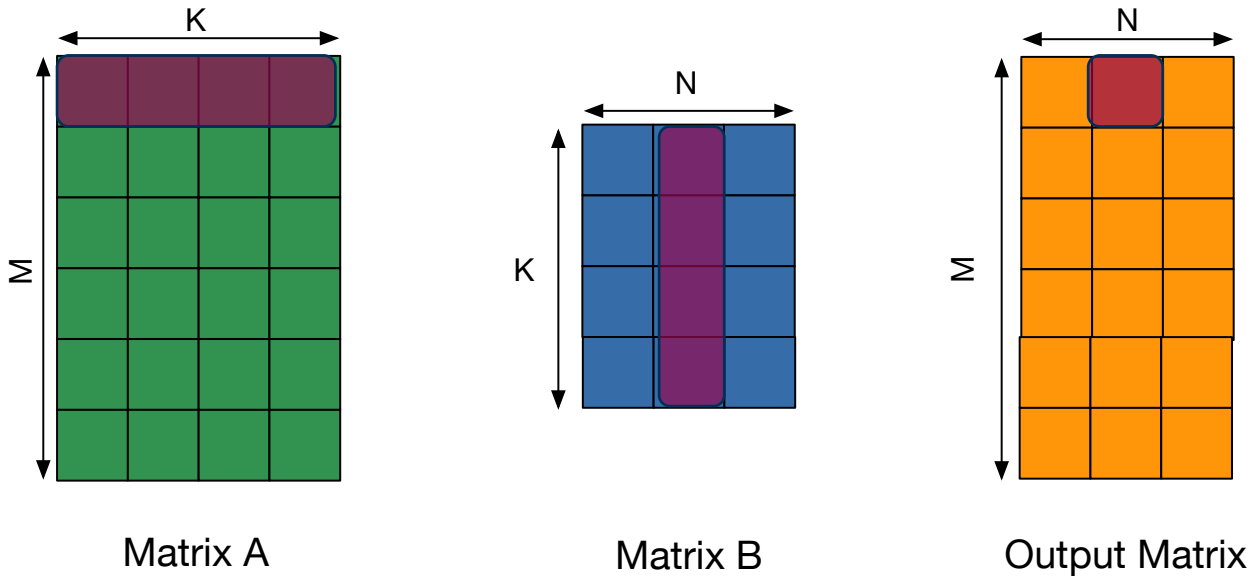
Matrix Multiplication

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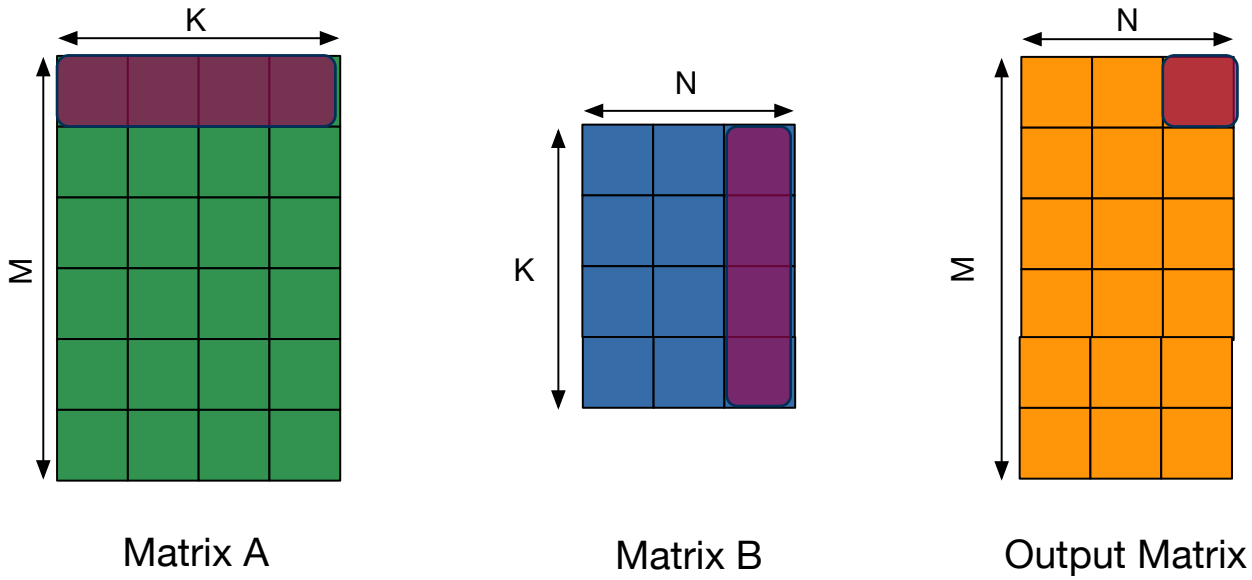


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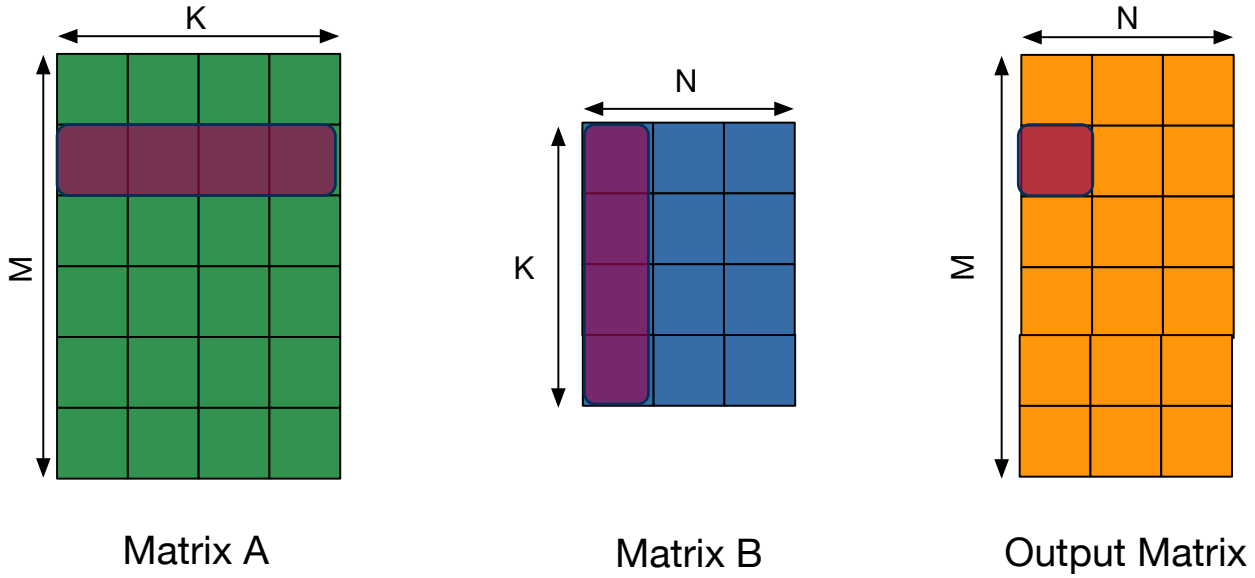


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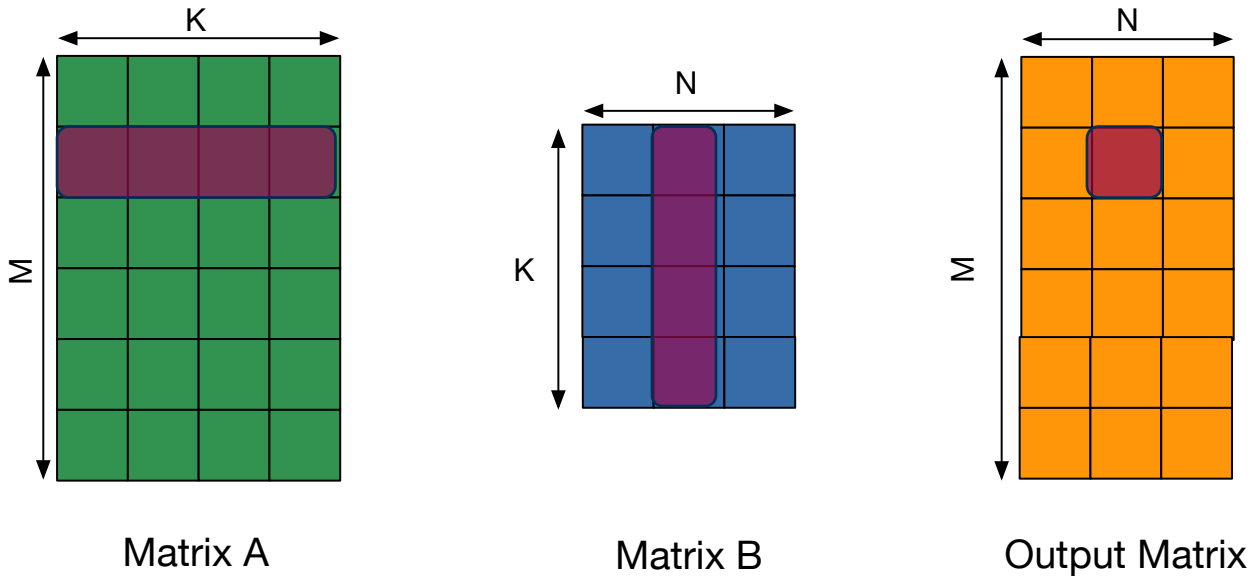


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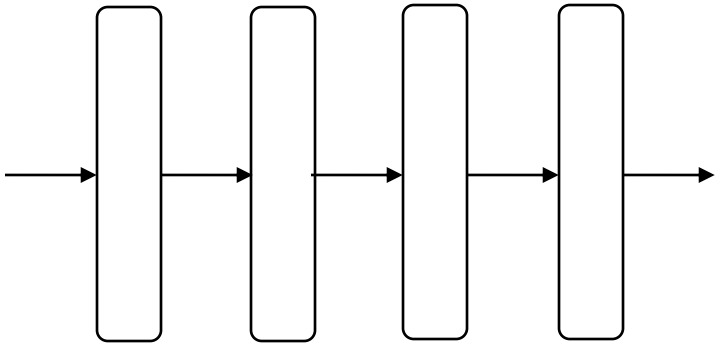
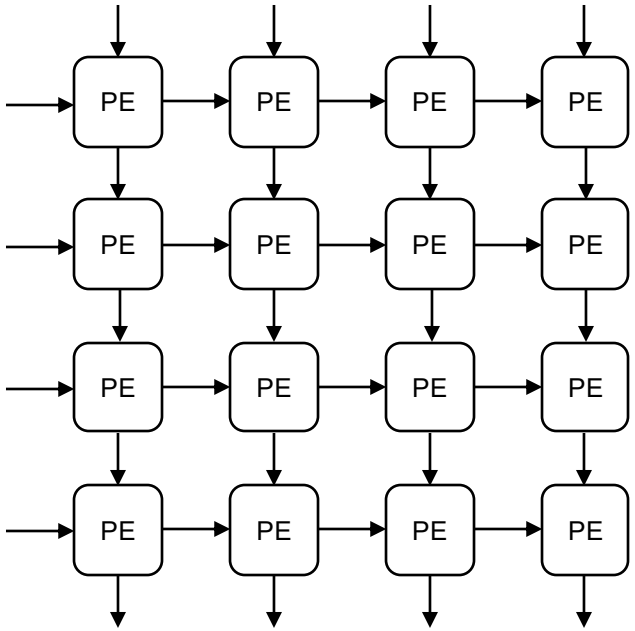
Systolic Arrays – 2D Pipeline



Traditional 1D Pipeline



2D Pipeline

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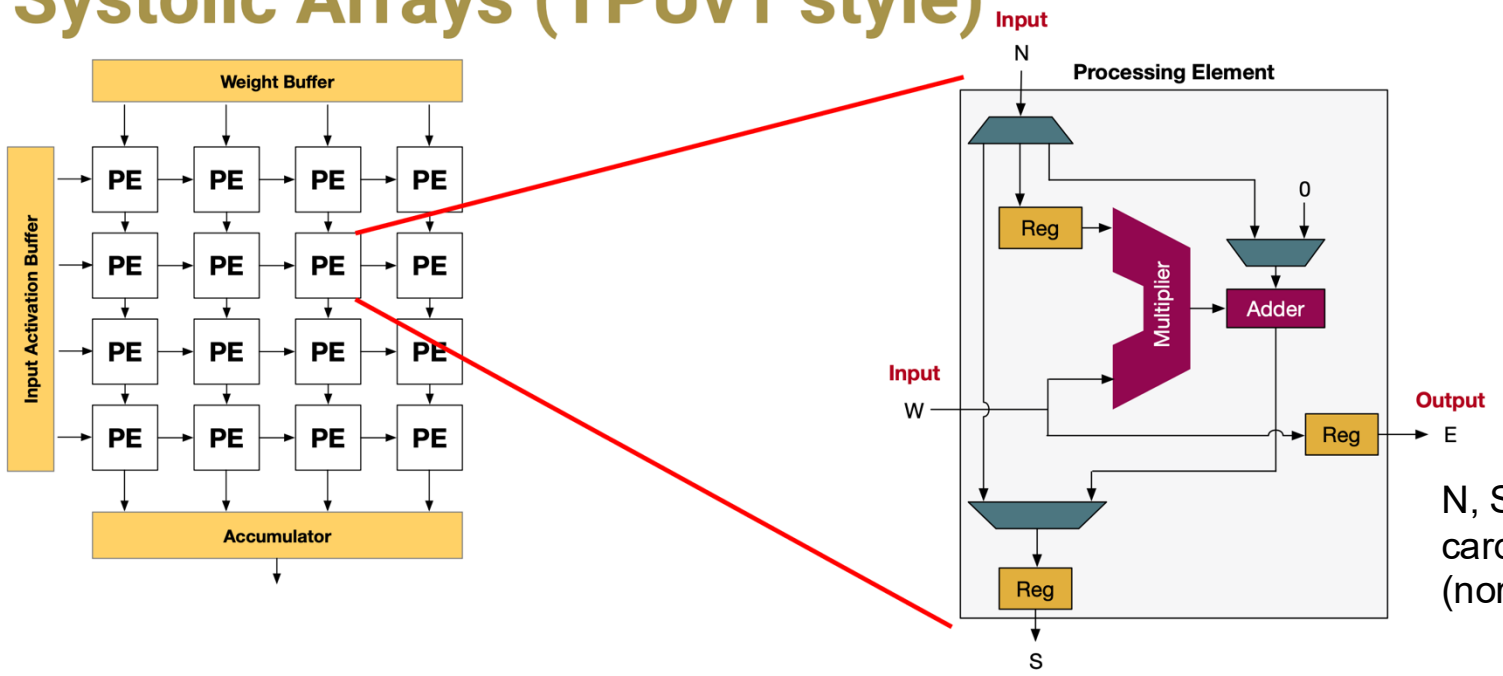
Systolic Arrays – Processing Elements (PEs)

- Simple PE design
  - No local scratchpad – global buffers are accessed through memory controllers
  - Registers store and forward weight, inputs, and outputs

PEs in Systolic Arrays (TPUv1 style)



N, S, E, W are cardinal directions (north, south, …)

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Systolic Arrays – Mapping

- There's multiple ways to map a GEneral Matrix Multiplication (GEMM) operation onto a 2D systolic array
  - Input Stationary
  - **Weight Stationary**
  - Output Stationary
- Determines the "flow" of data across the array
  - Temporal vs Spatial re-use of data
  - Spatial dimensions are streamed in/out
  - Temporal dimensions remain for the entirety of the computation

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Systolic Arrays – Weight Stationary Mapping

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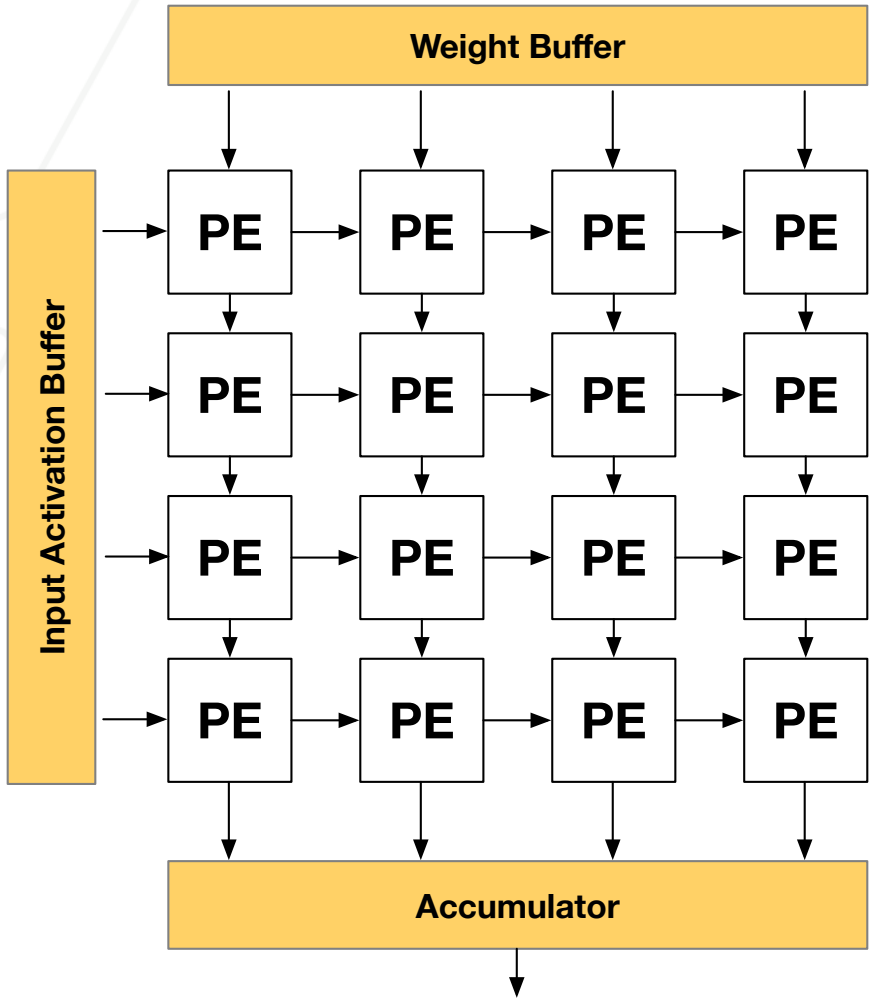
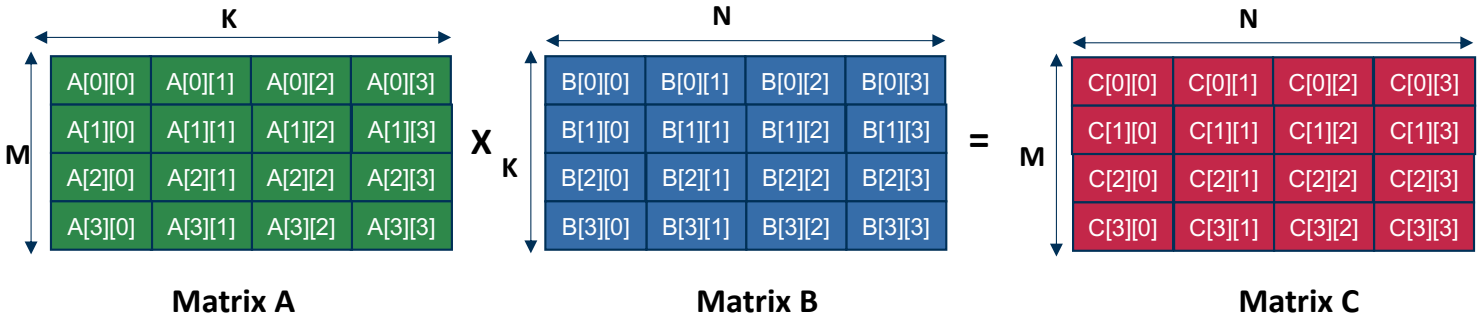
Matrix Multiplication on a Systolic Array



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for m in range(4):
  for n in range(4):
    for k in range(4):
      C[m][n] += A[m][k] * B[k][n]
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Assume following "mapping"

- Column: Reduced (*aka* contracted) Dimension (K)
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Note: alternate mapping styles possible

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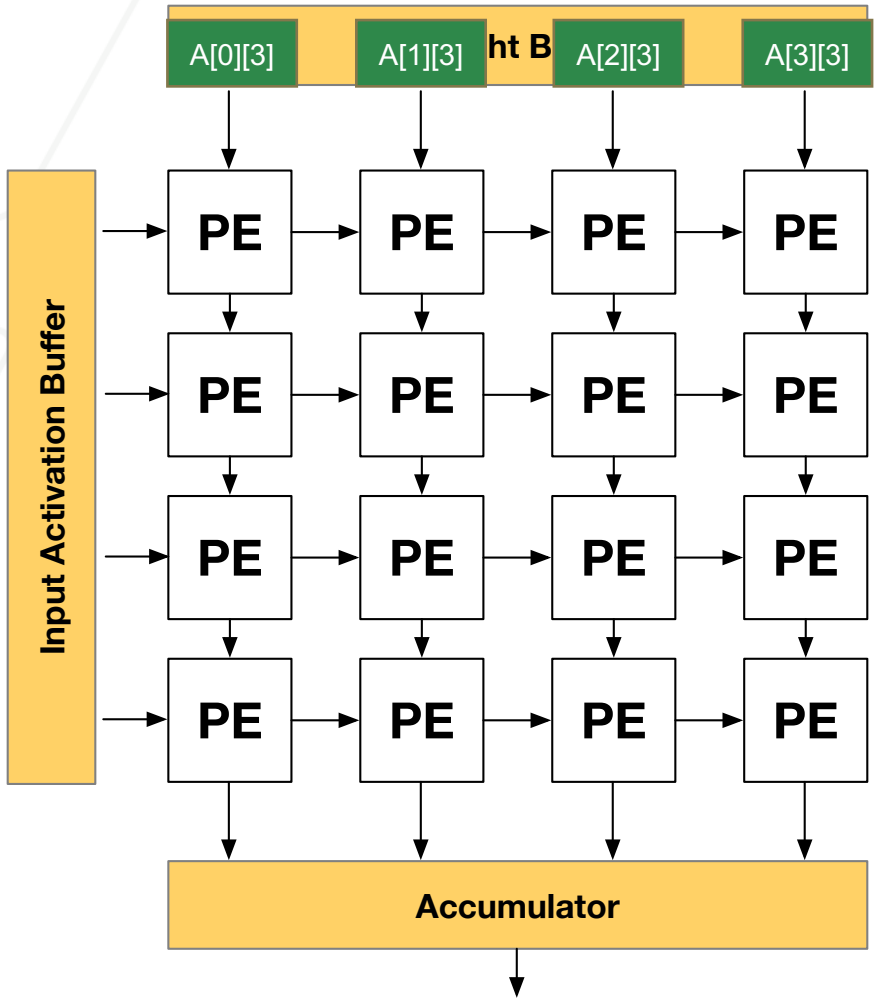
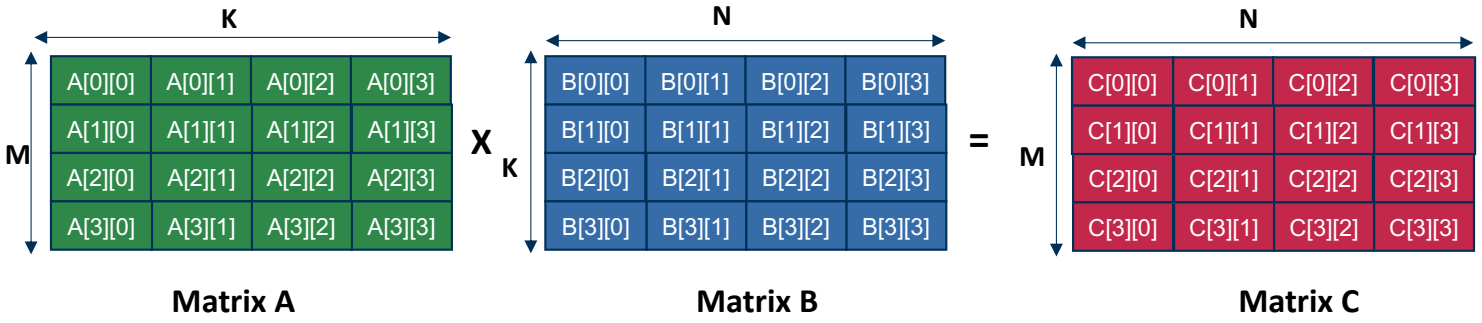
Matrix Multiplication on a Systolic Array

Phase 1:
Load Weight Tensor



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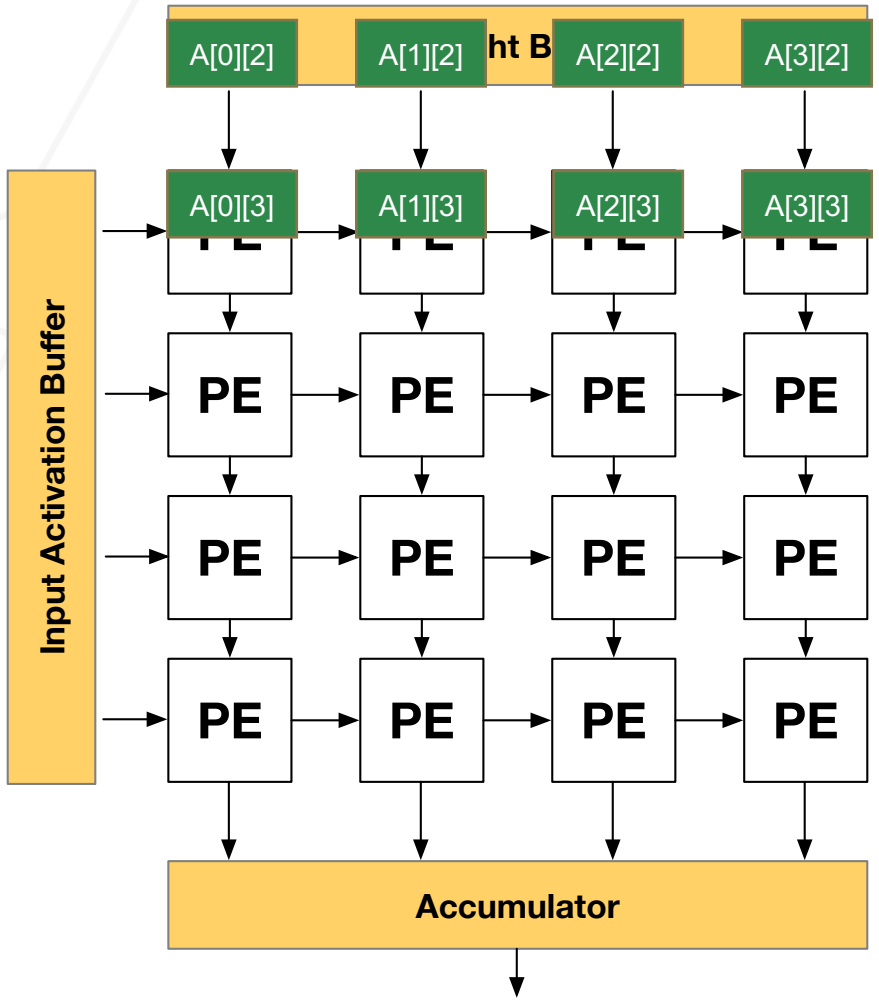

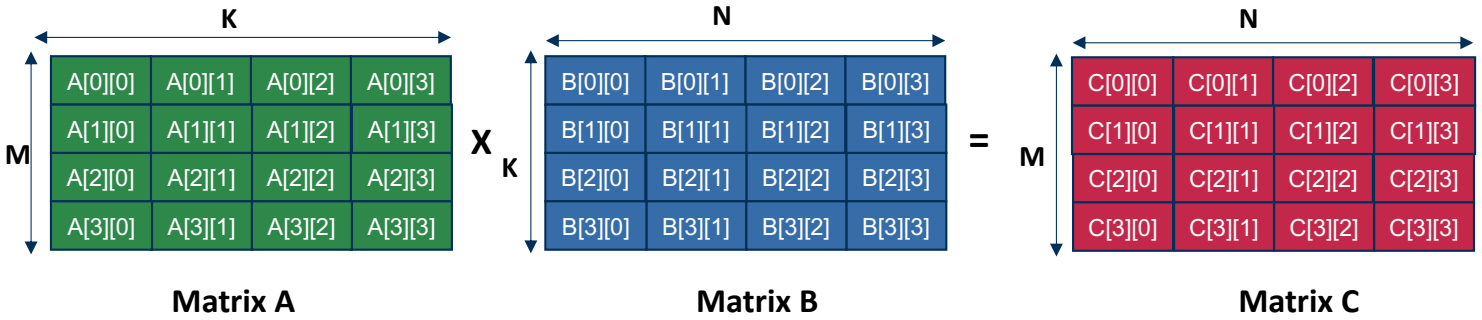


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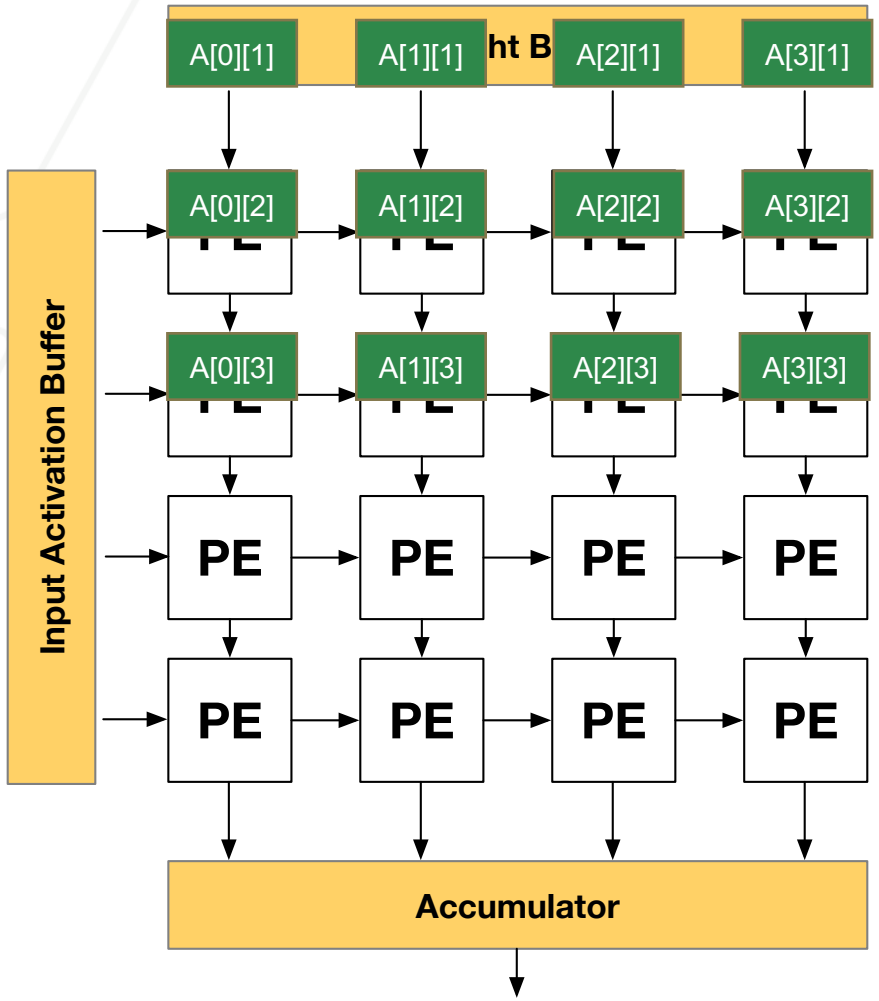

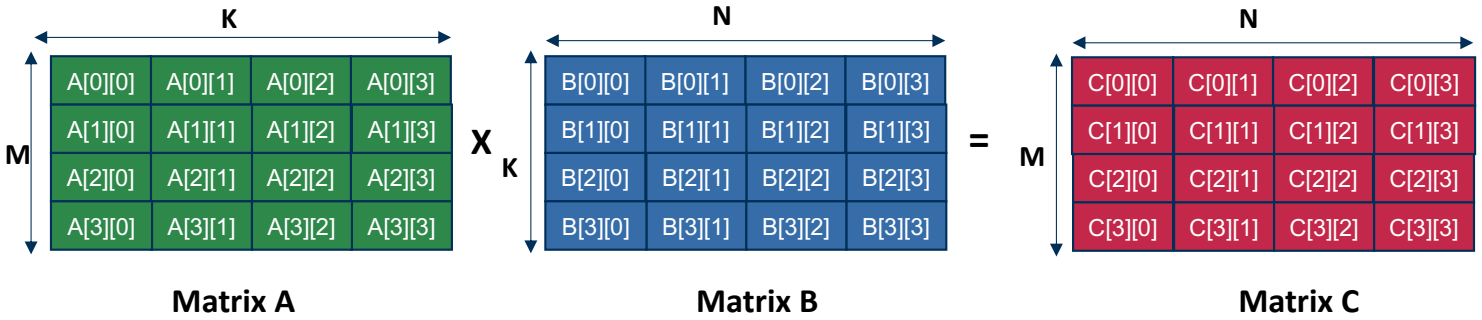


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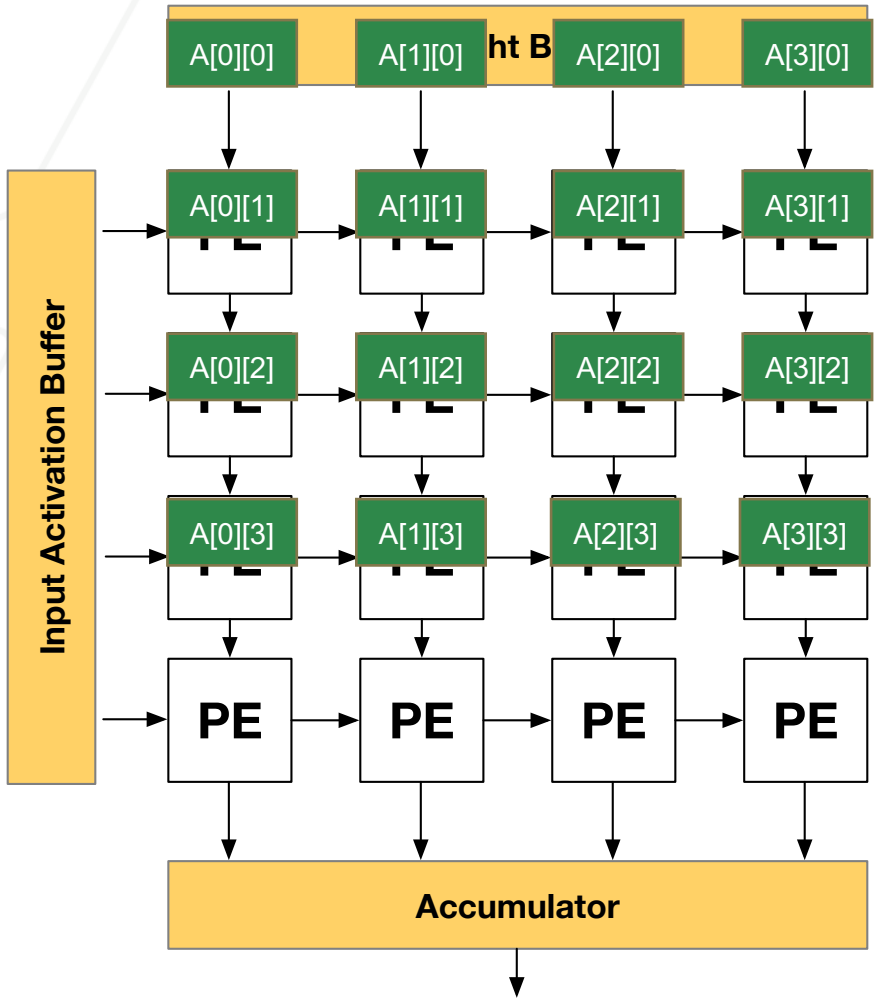

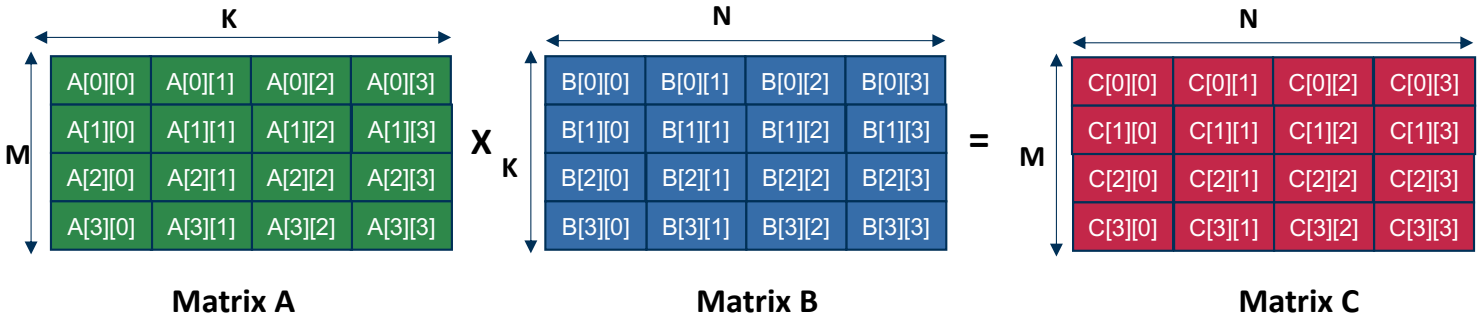


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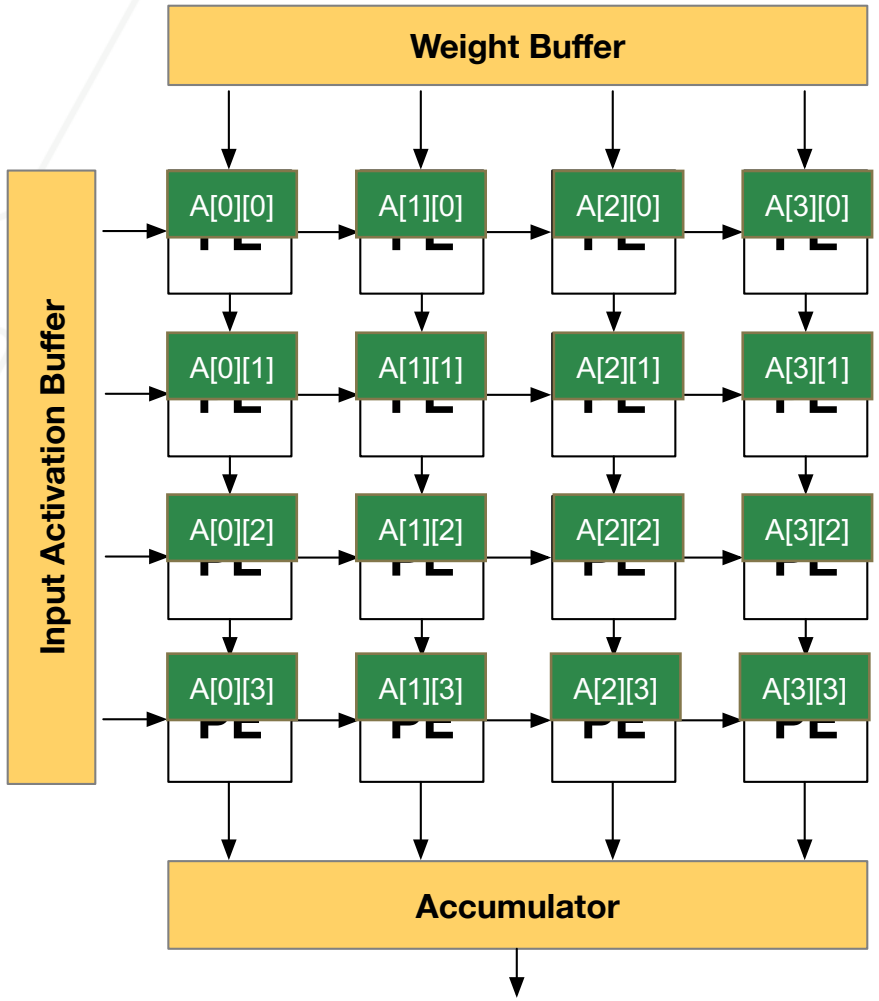

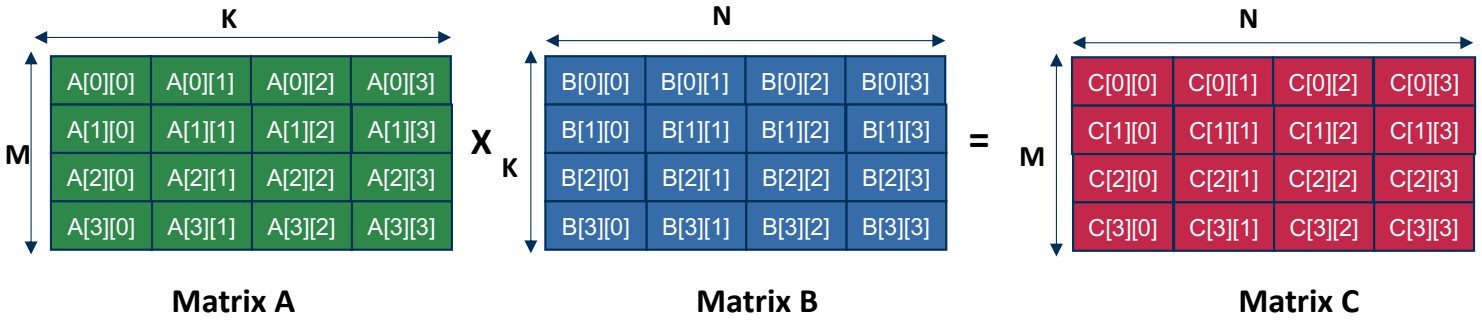


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Matrix A will now stay resident (i.e., "stationary") in the PEs

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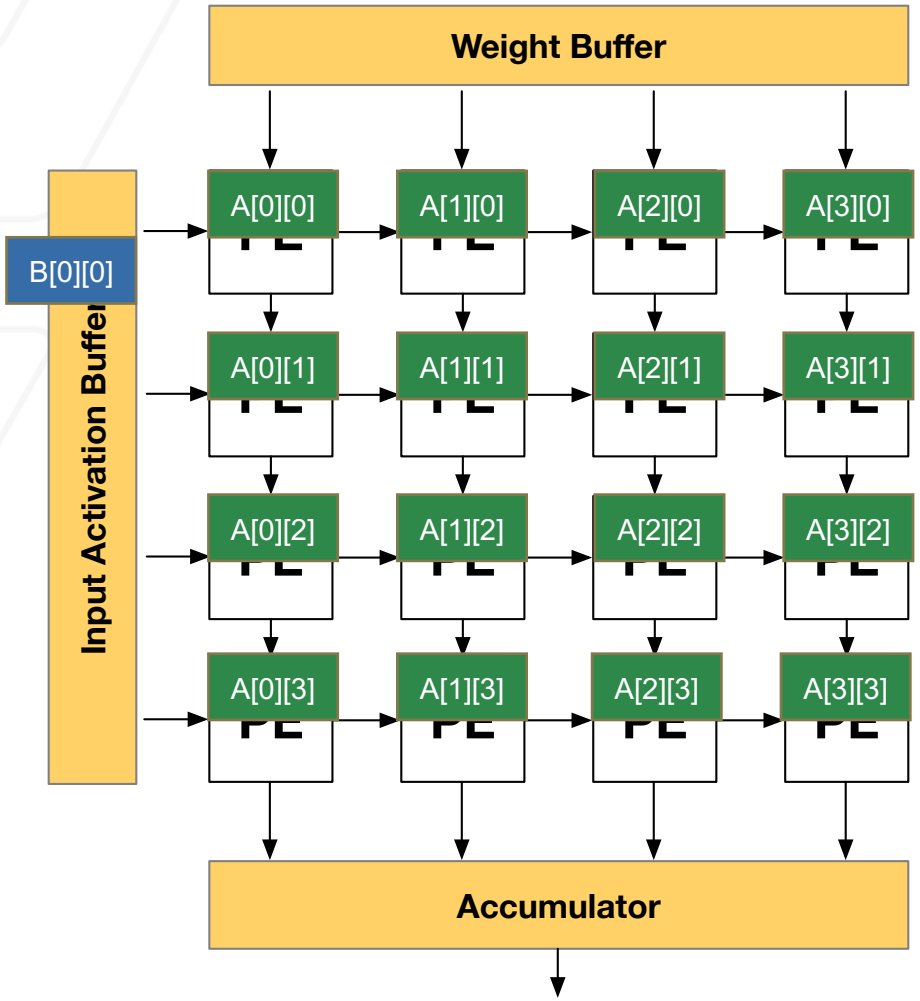
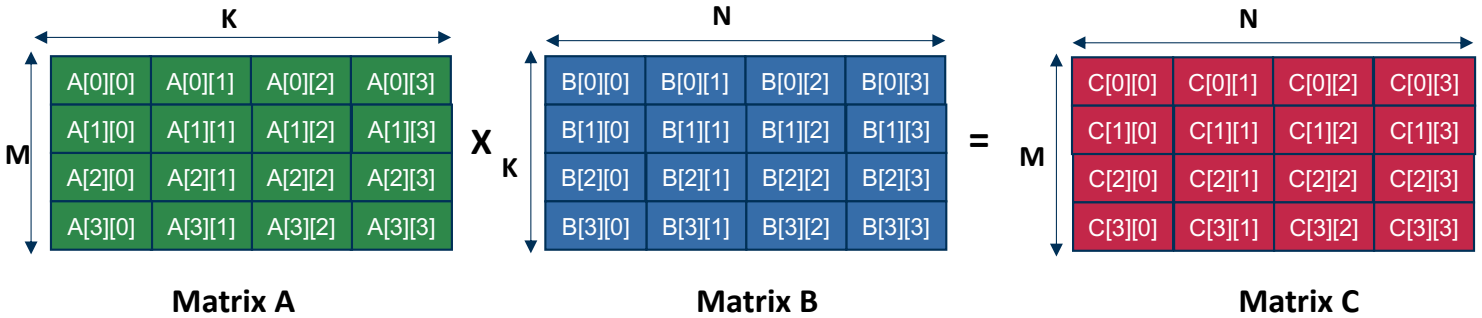
Matrix Multiplication on a Systolic Array

Phase 2:
Stream Activation Tensor



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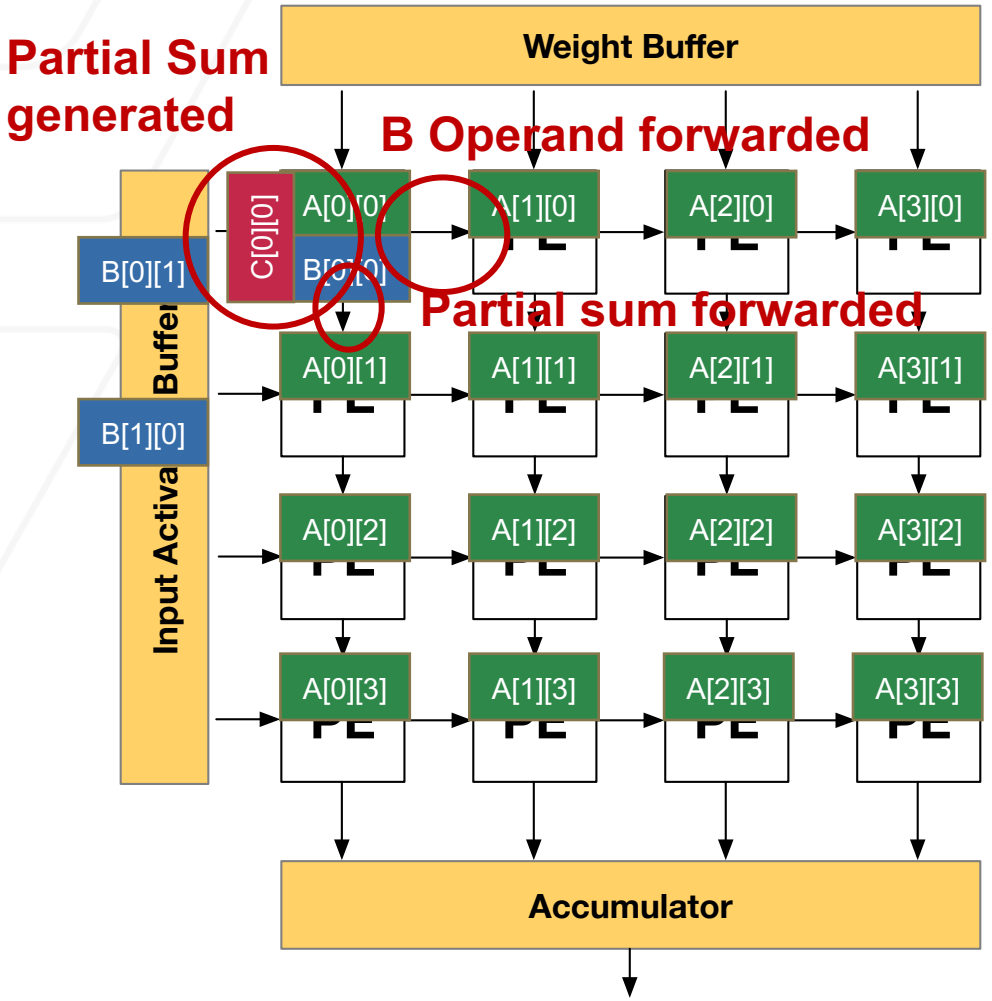

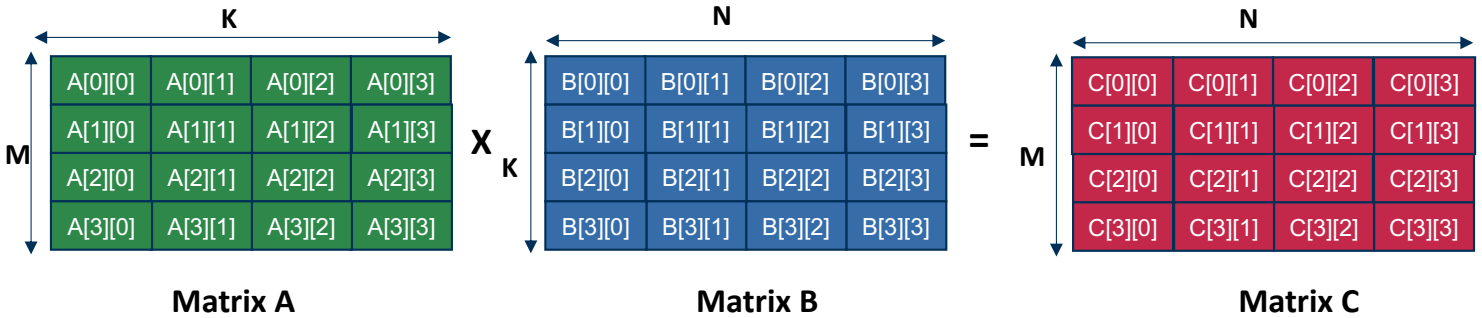


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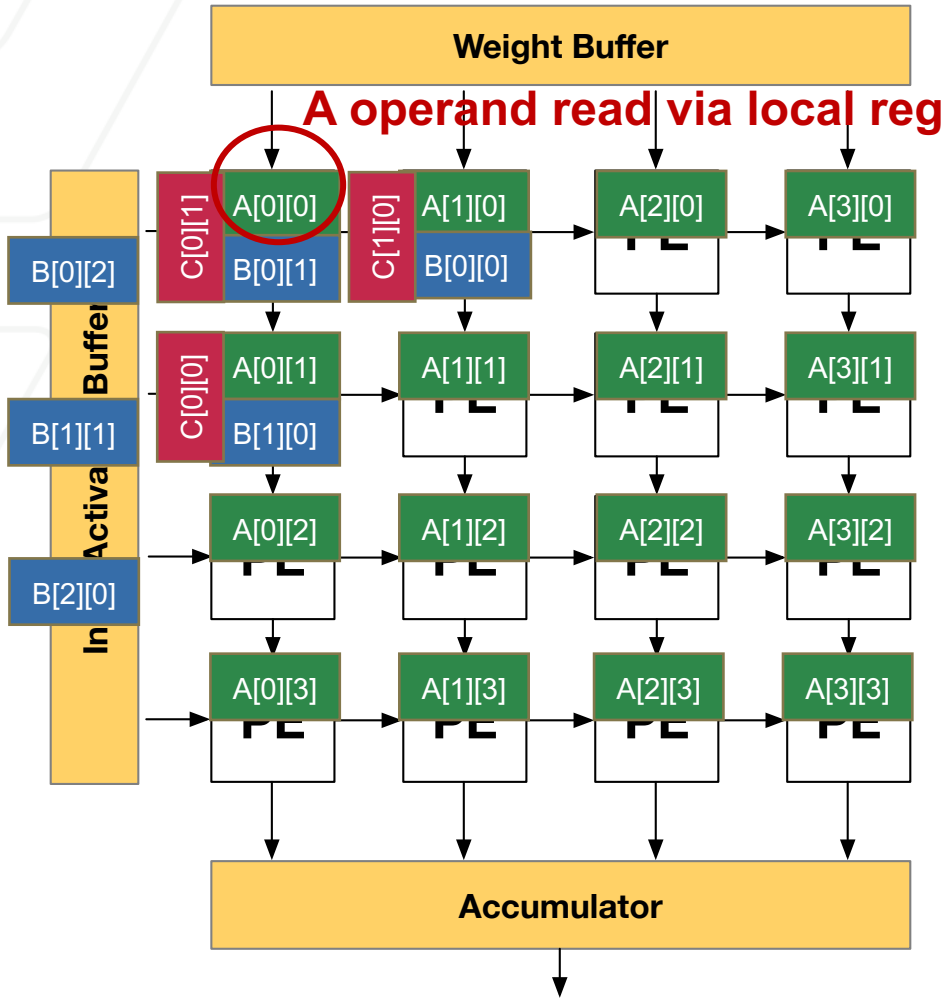

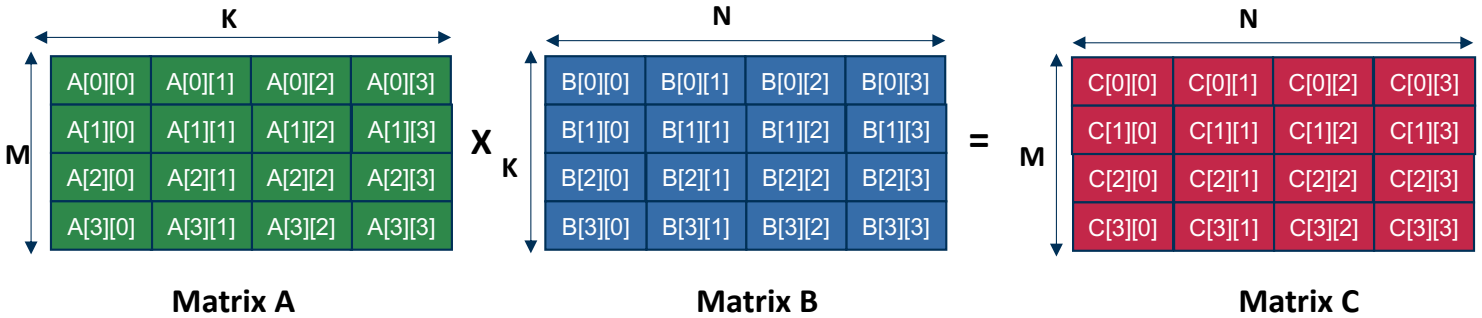


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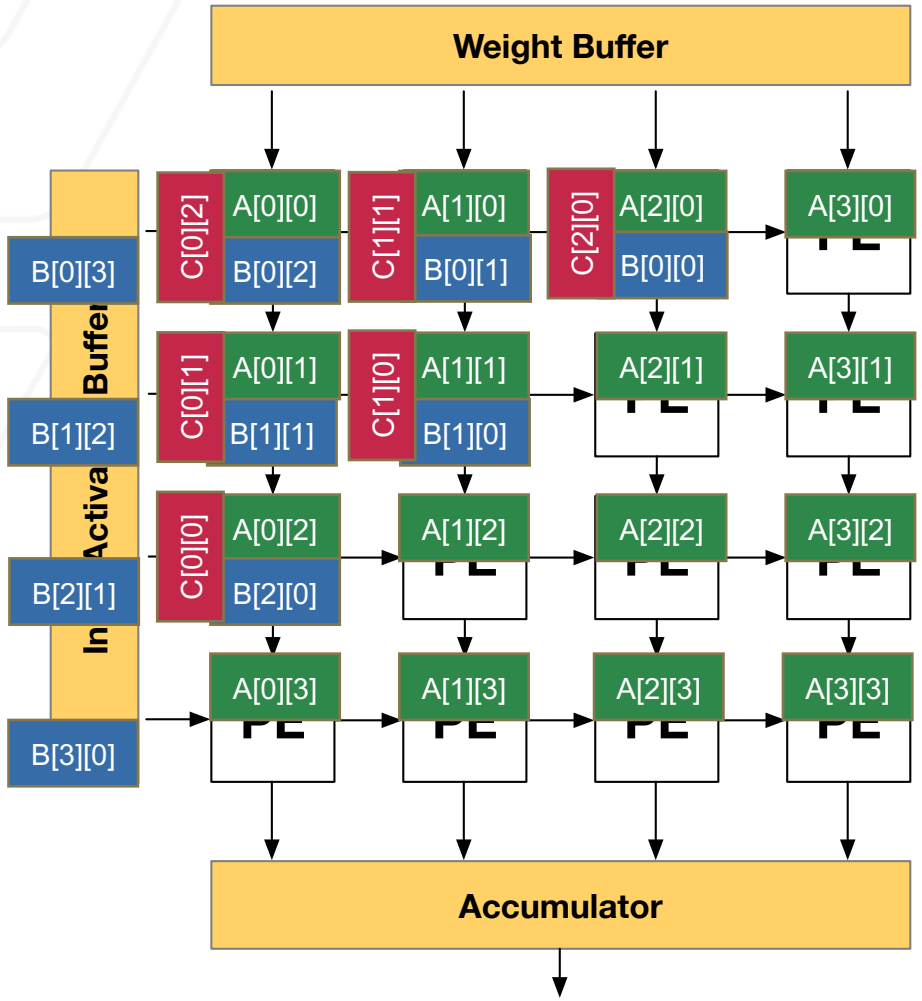

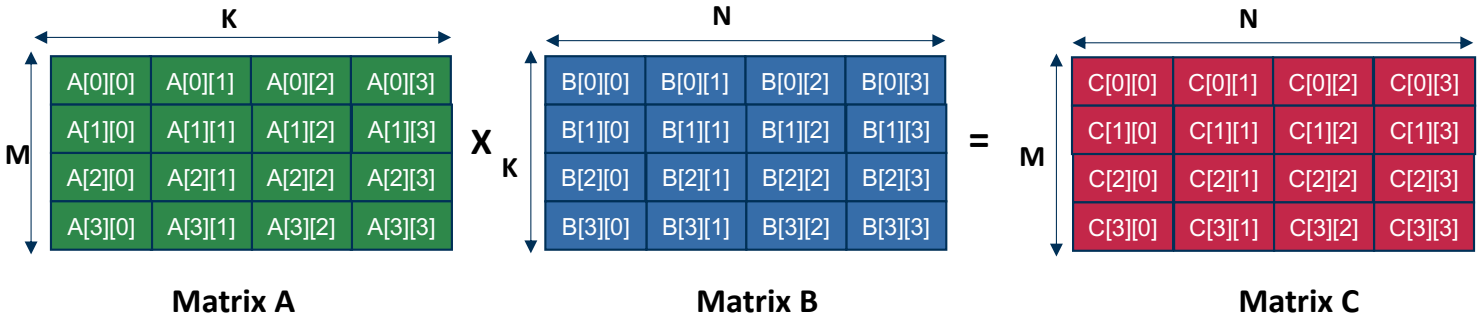


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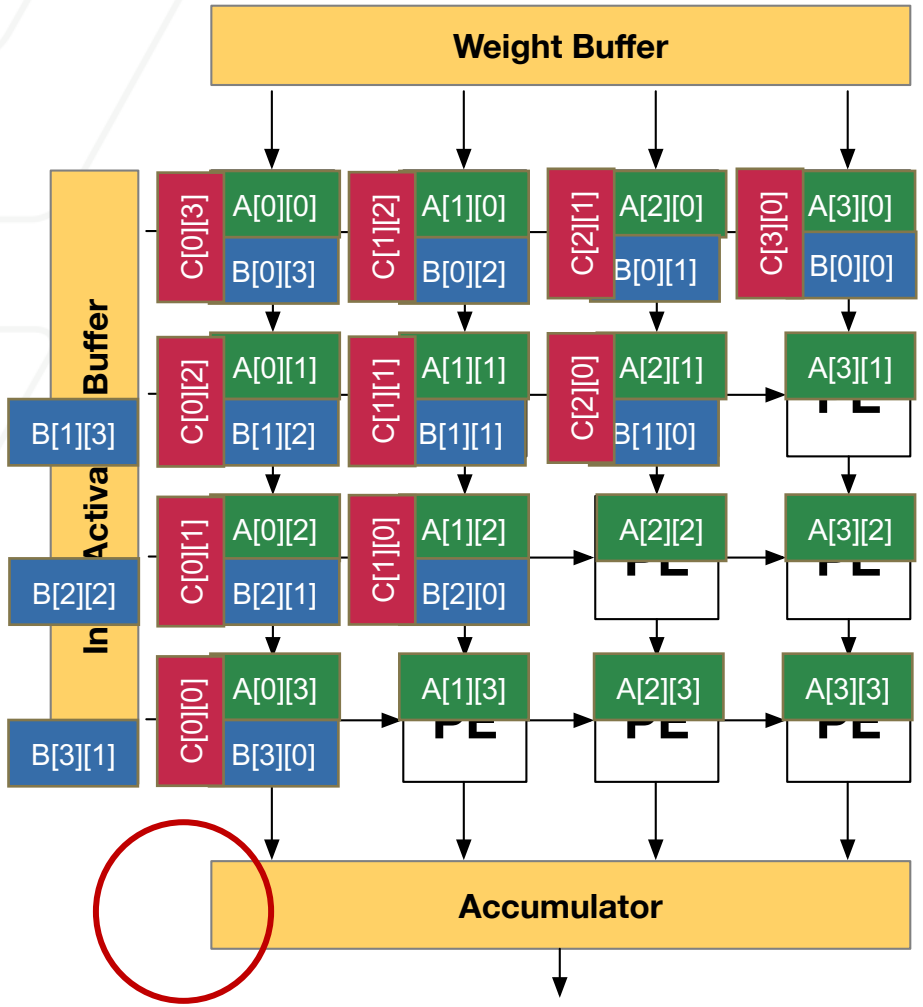
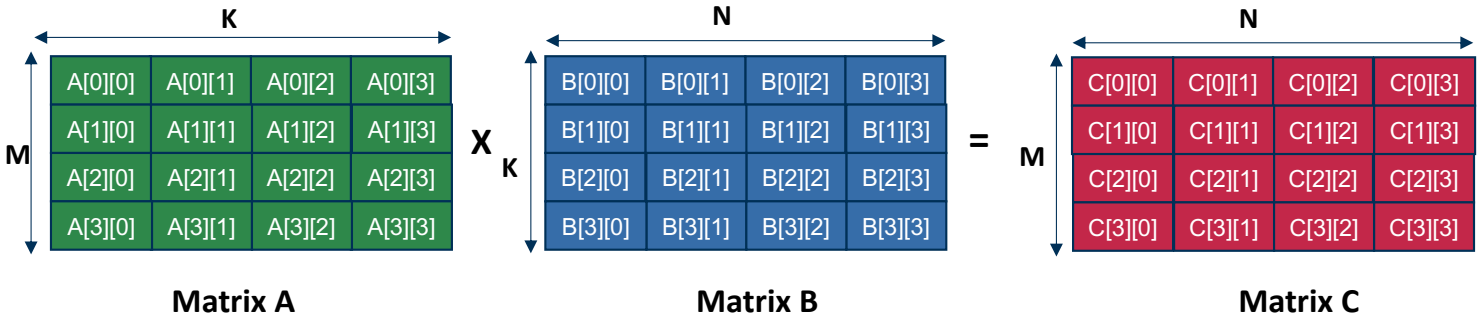


First output generated

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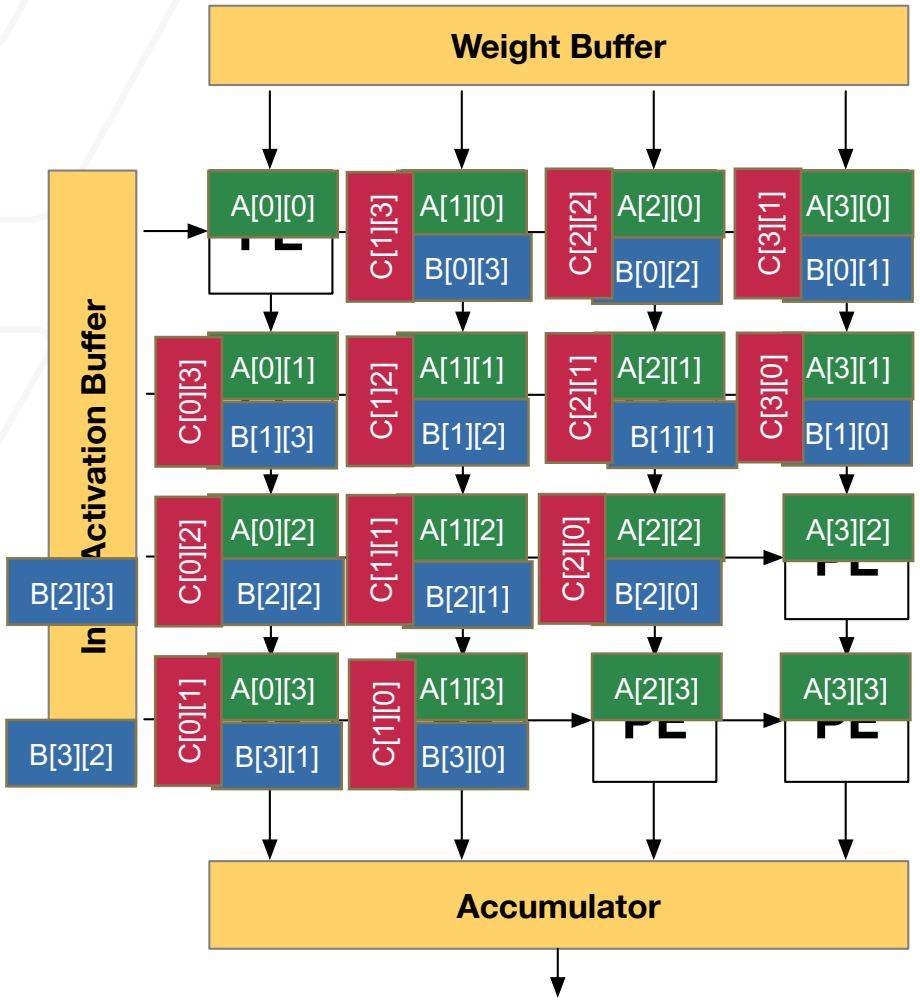

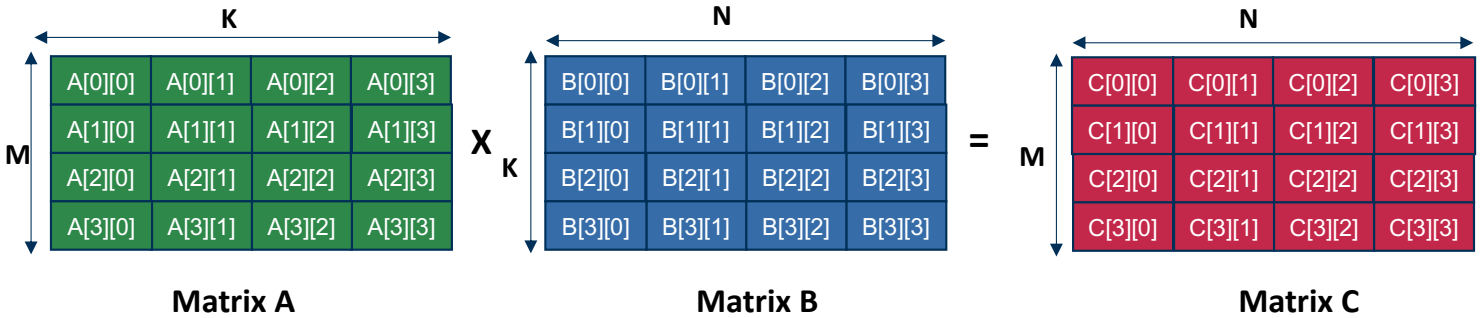


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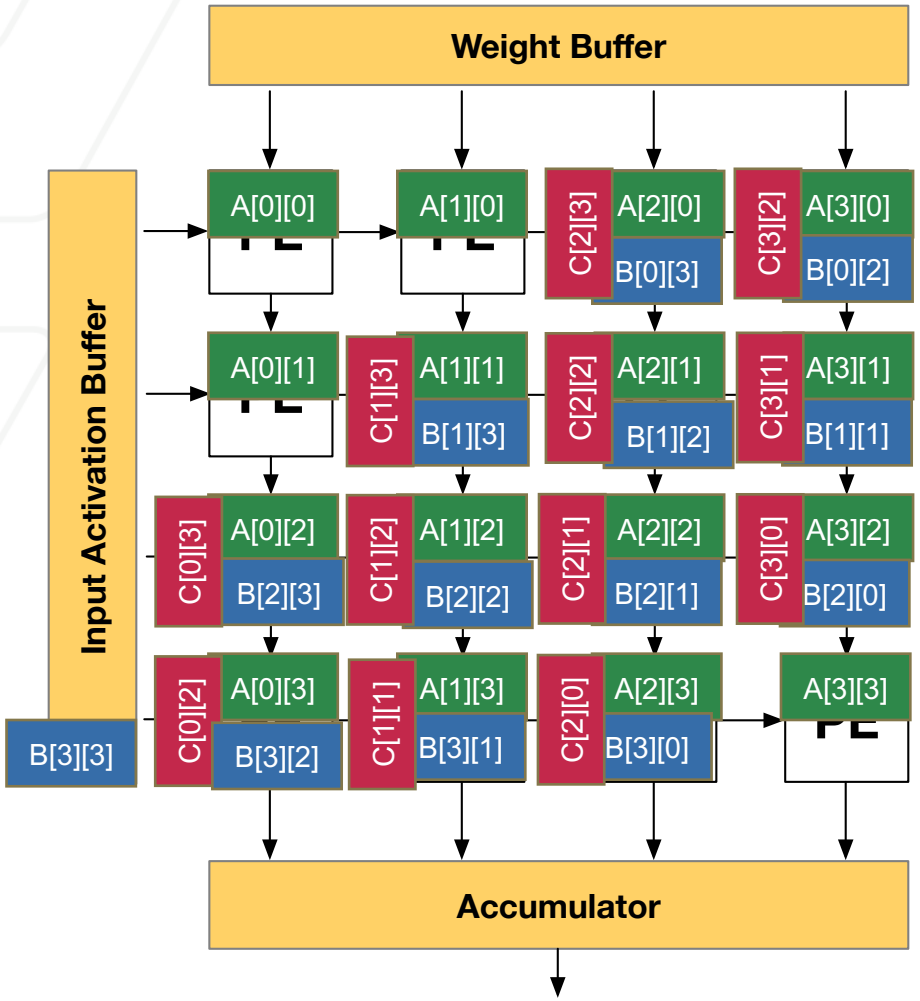

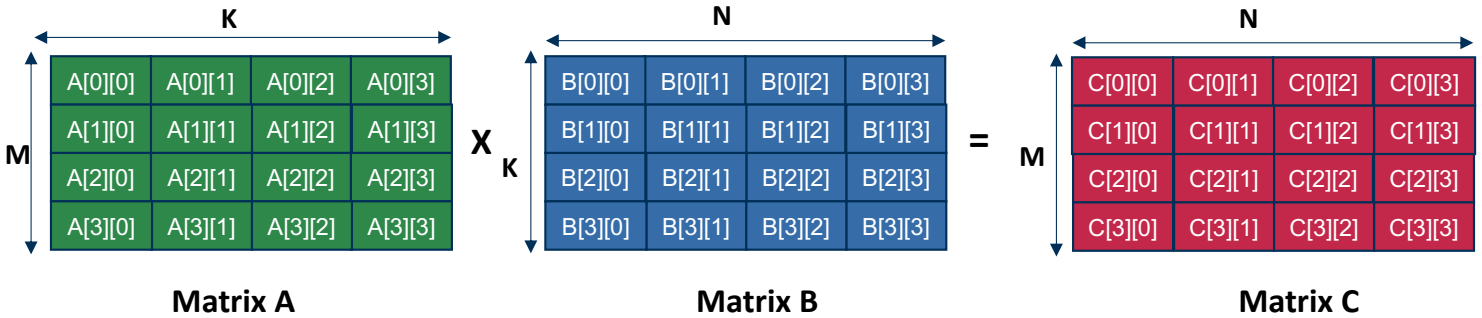


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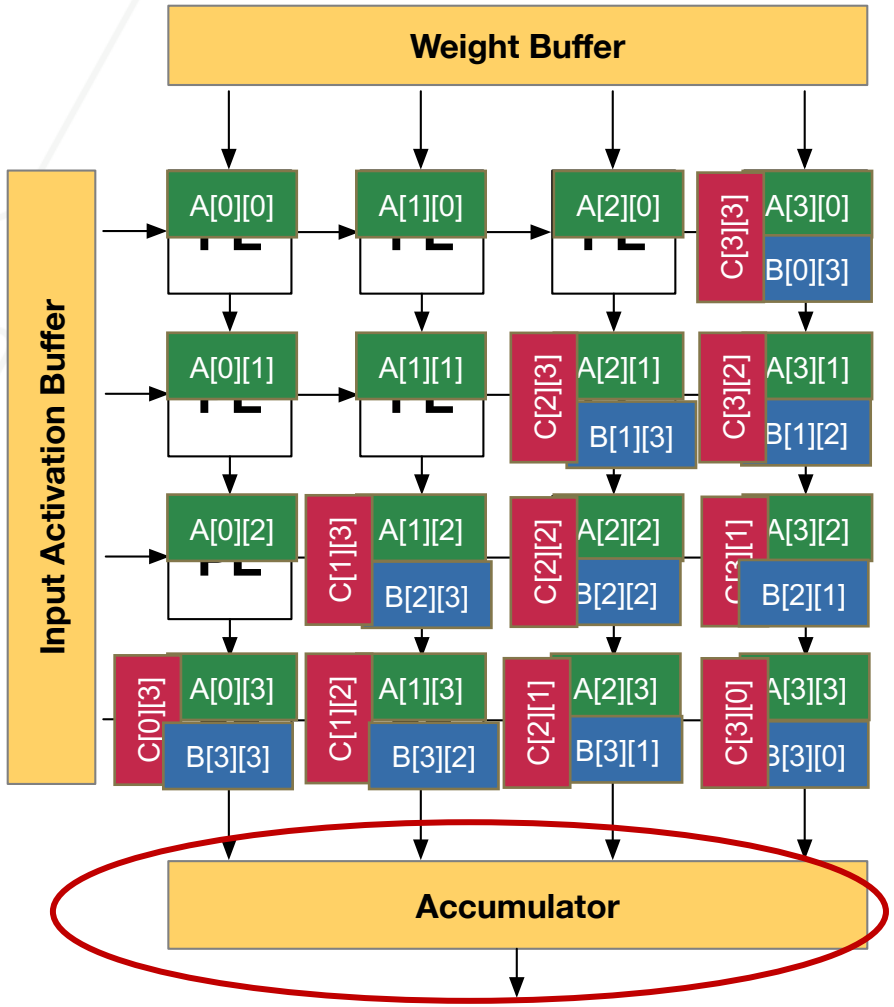
Note: alternate mapping styles possible

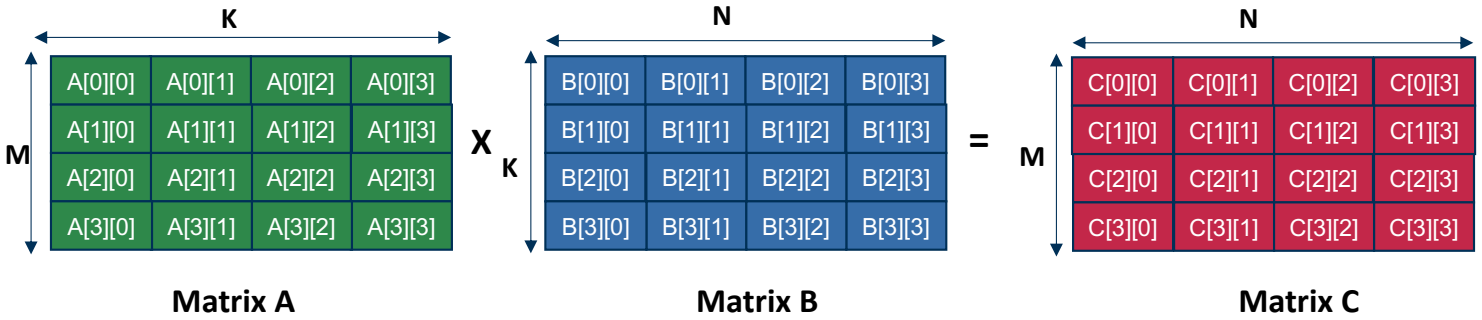
Slide Courtesy: Dr. Tushar Krishna (GT ECE – Synergy Lab)

Matrix Multiplication on a Systolic Array

Phase 2:
Stream Activation Tensor



Four outputs every cycle in steady state

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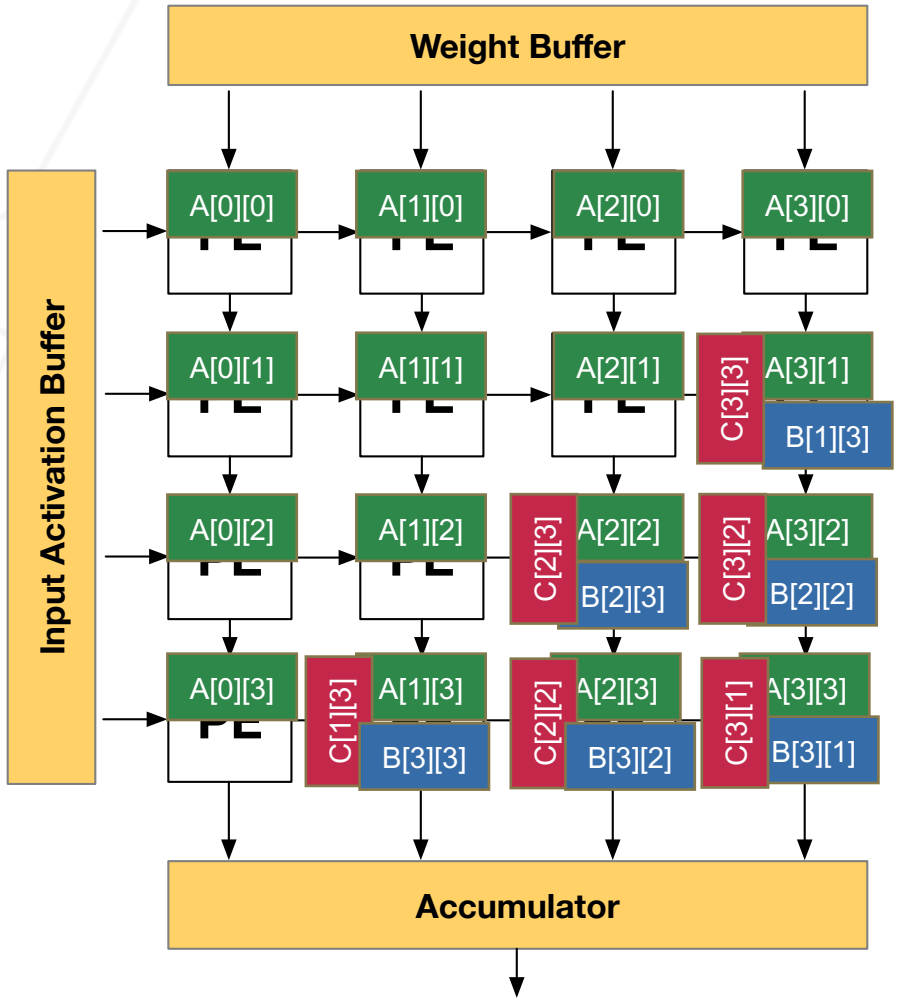

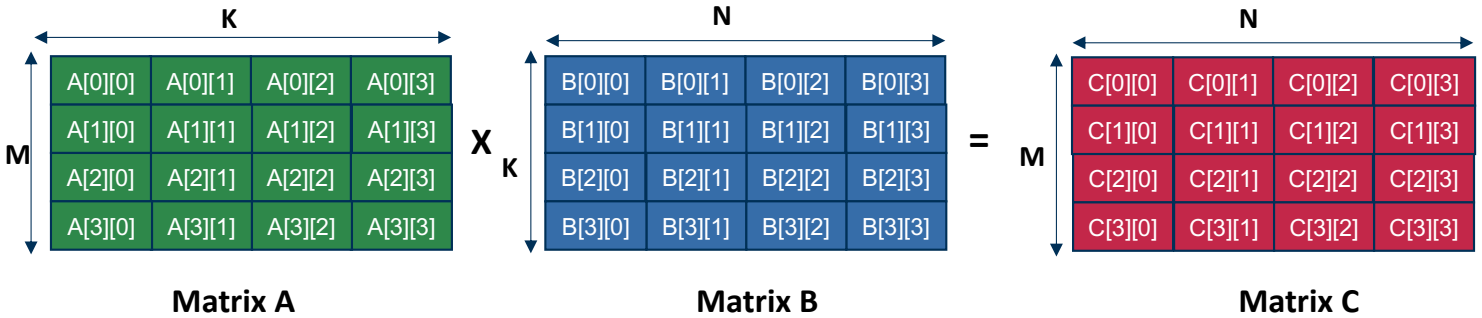


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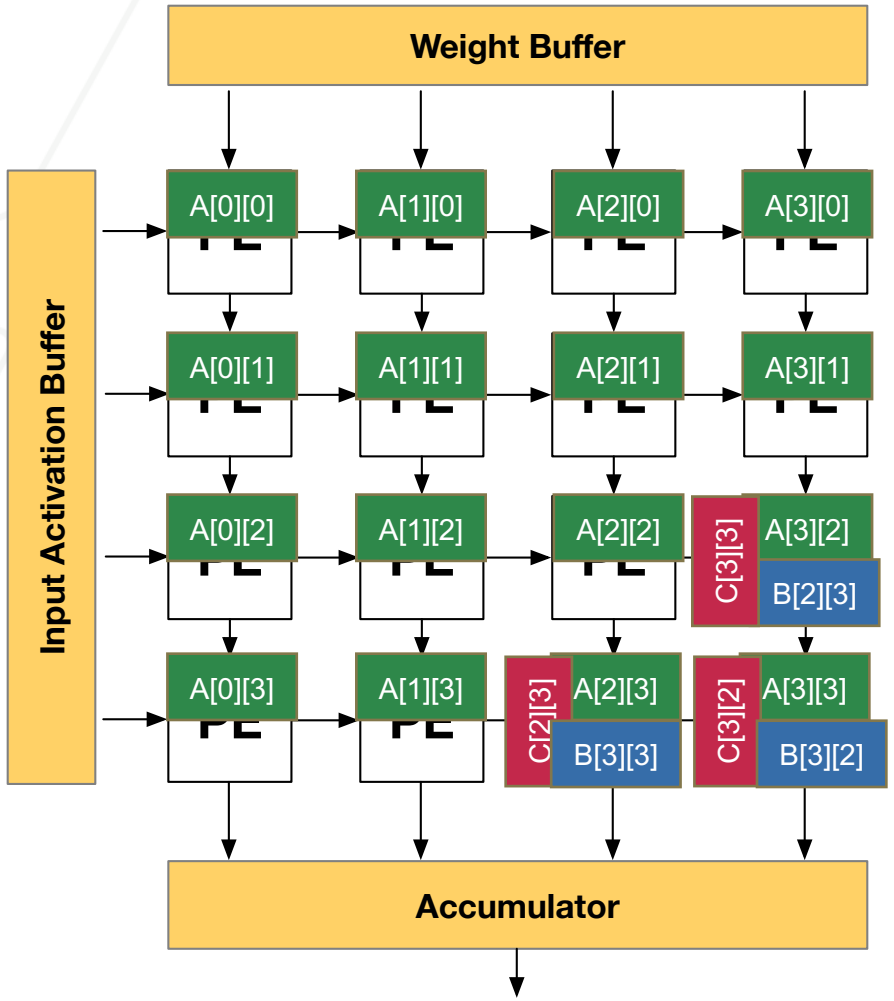

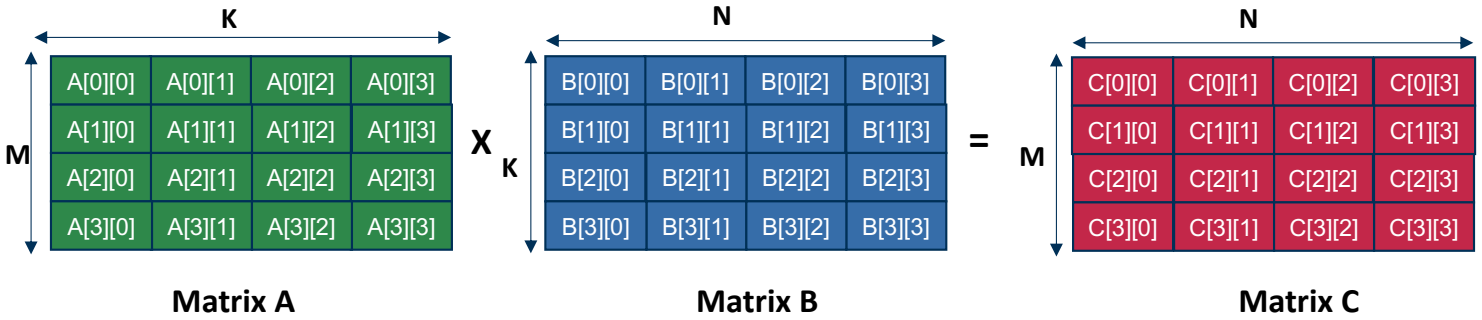


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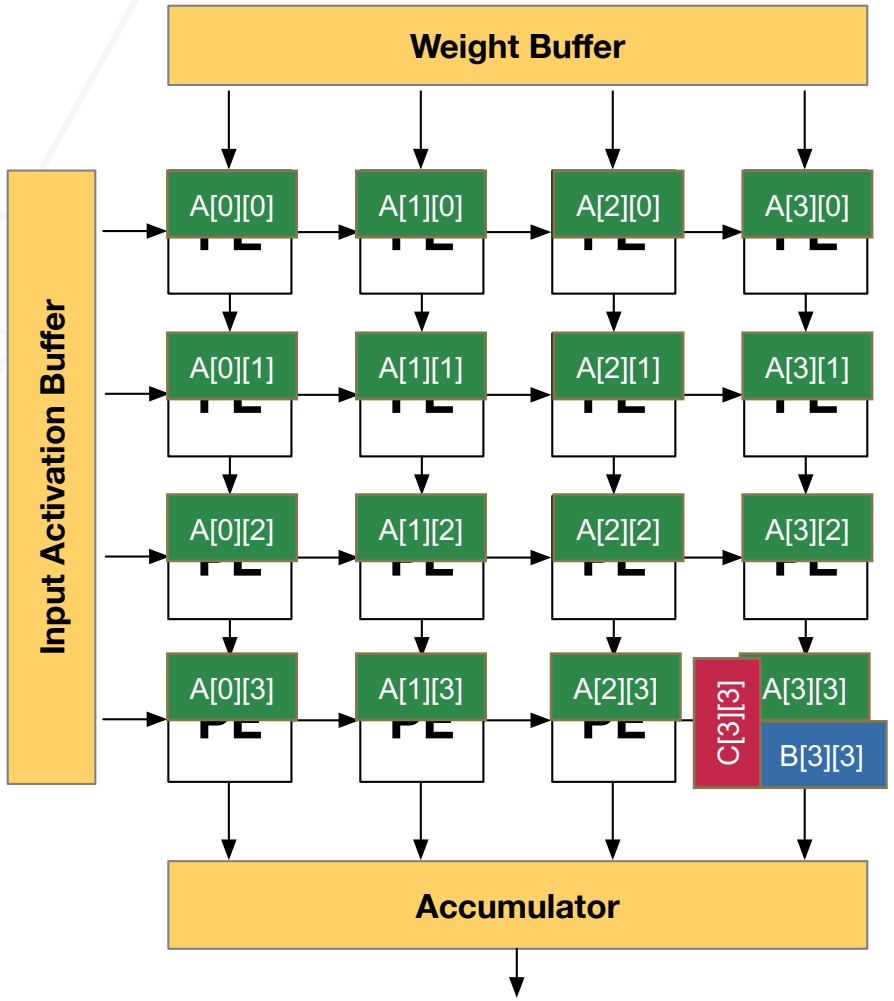

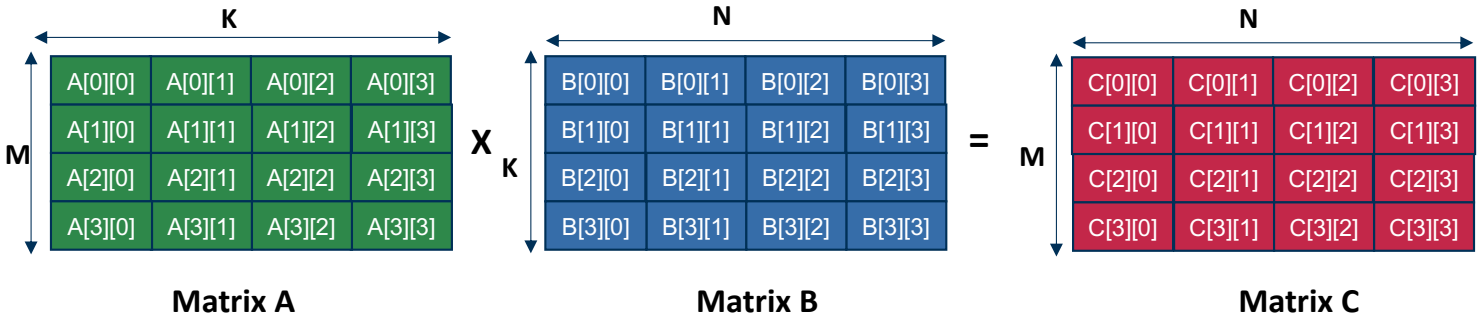


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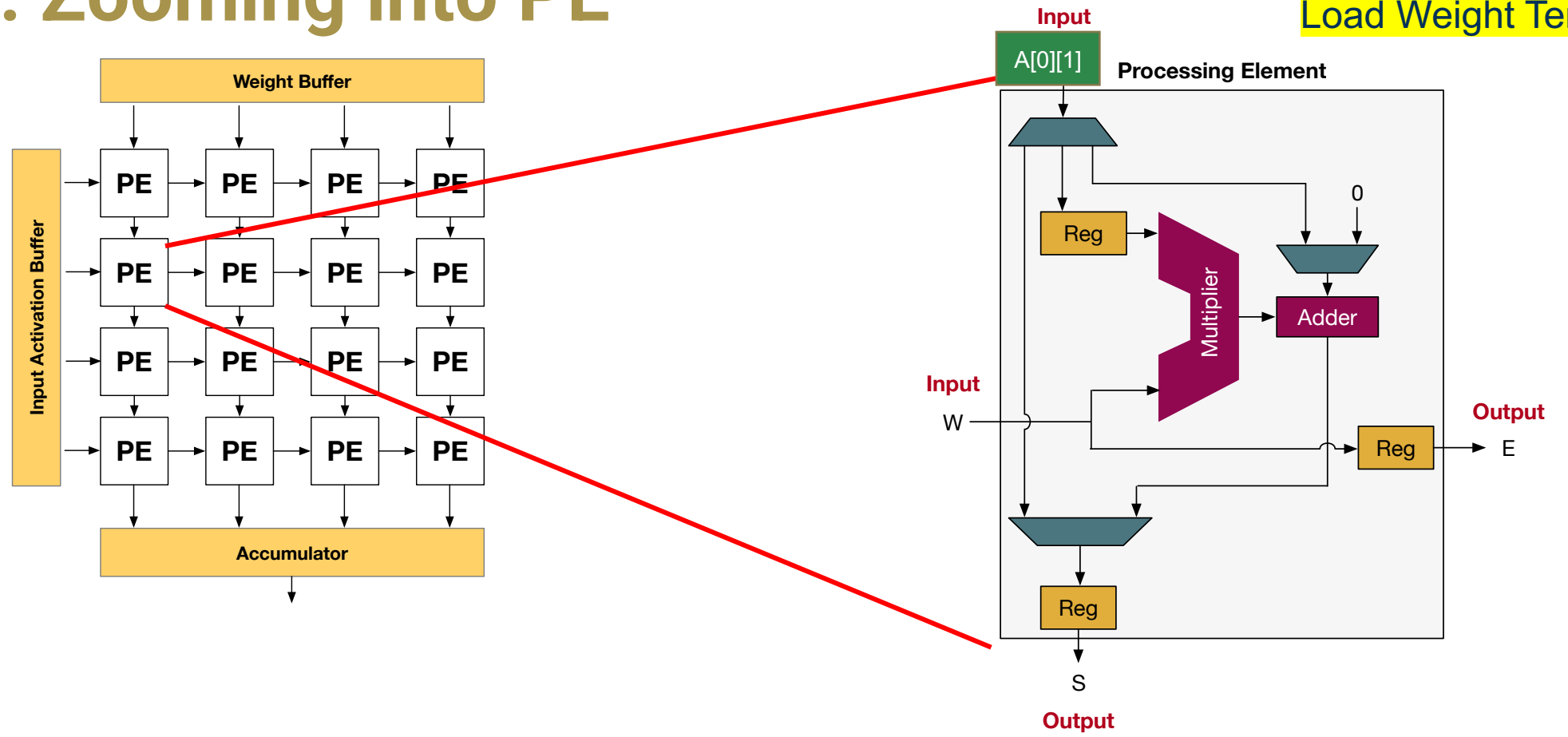
Recap: Zooming into PE

Phase 1:
Load Weight Tensor



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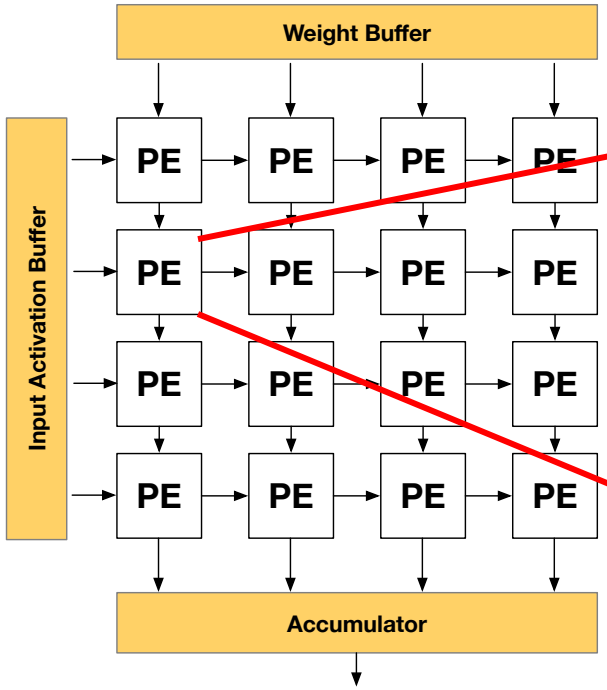

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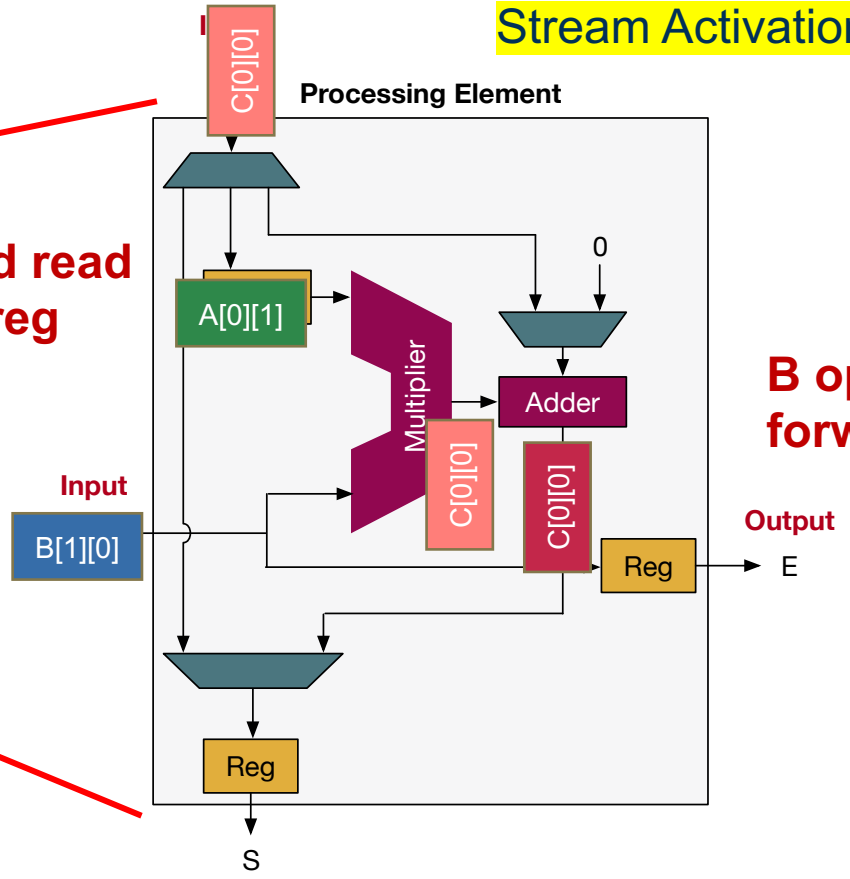


A operand read via local reg

B operand forwarded

Partial sum forwarded

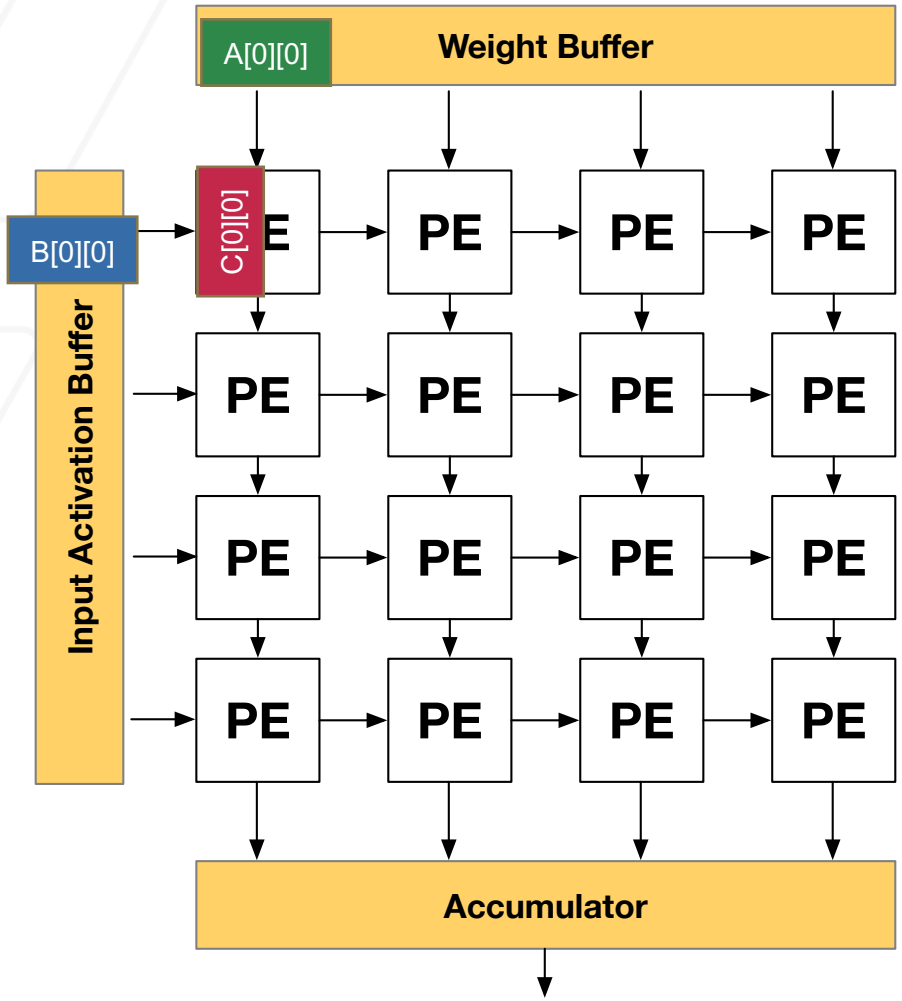
Slide Courtesy: Dr. Tushar Krishna (GT ECE – Synergy Lab)

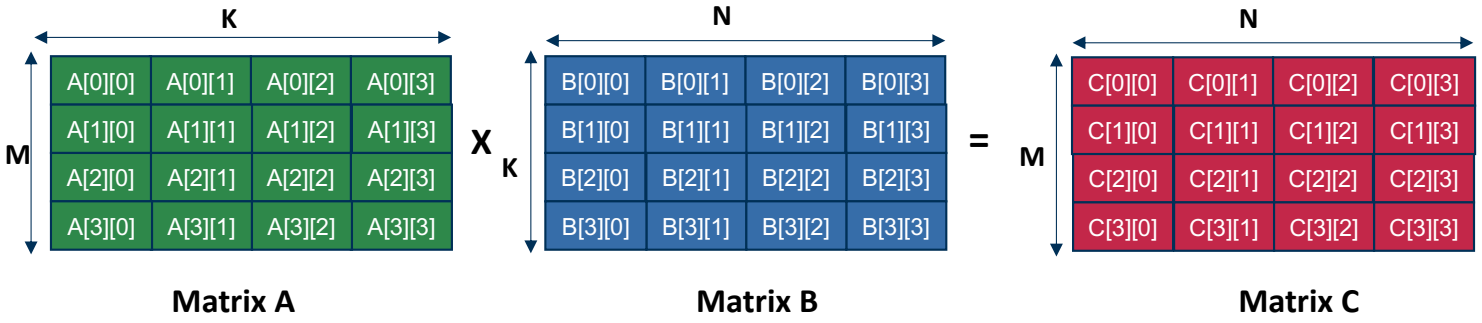
Systolic Arrays – Output Stationary Mapping

The following slides are courtesy of Dr. Tushar Krishna (GT ECE – Synergy Lab)

Georgia Tech

Matrix Multiplication on a Systolic Array



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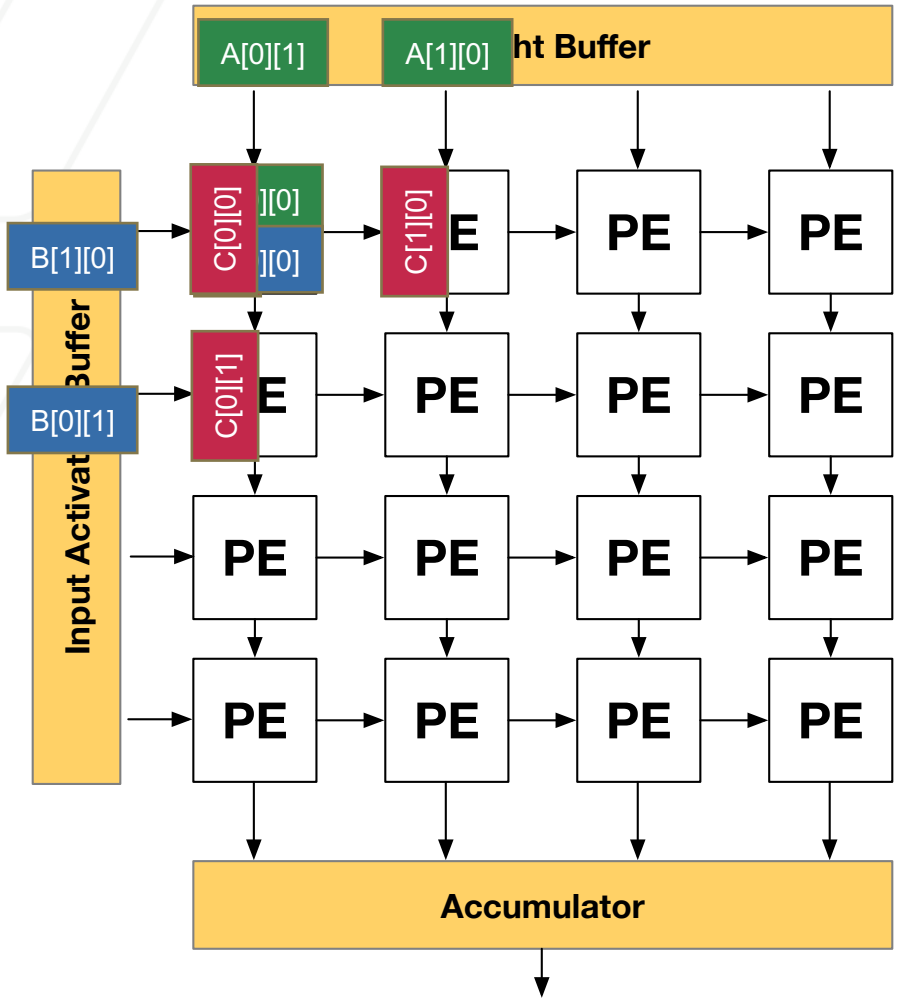

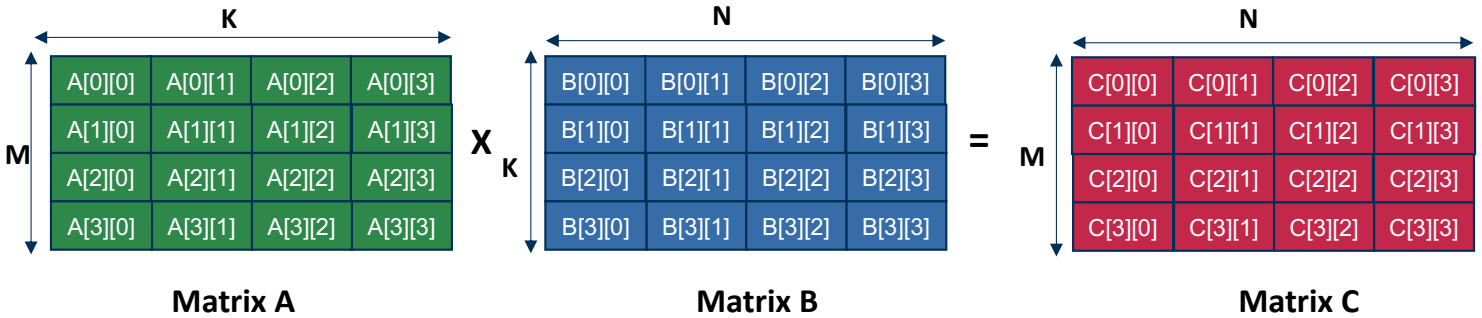


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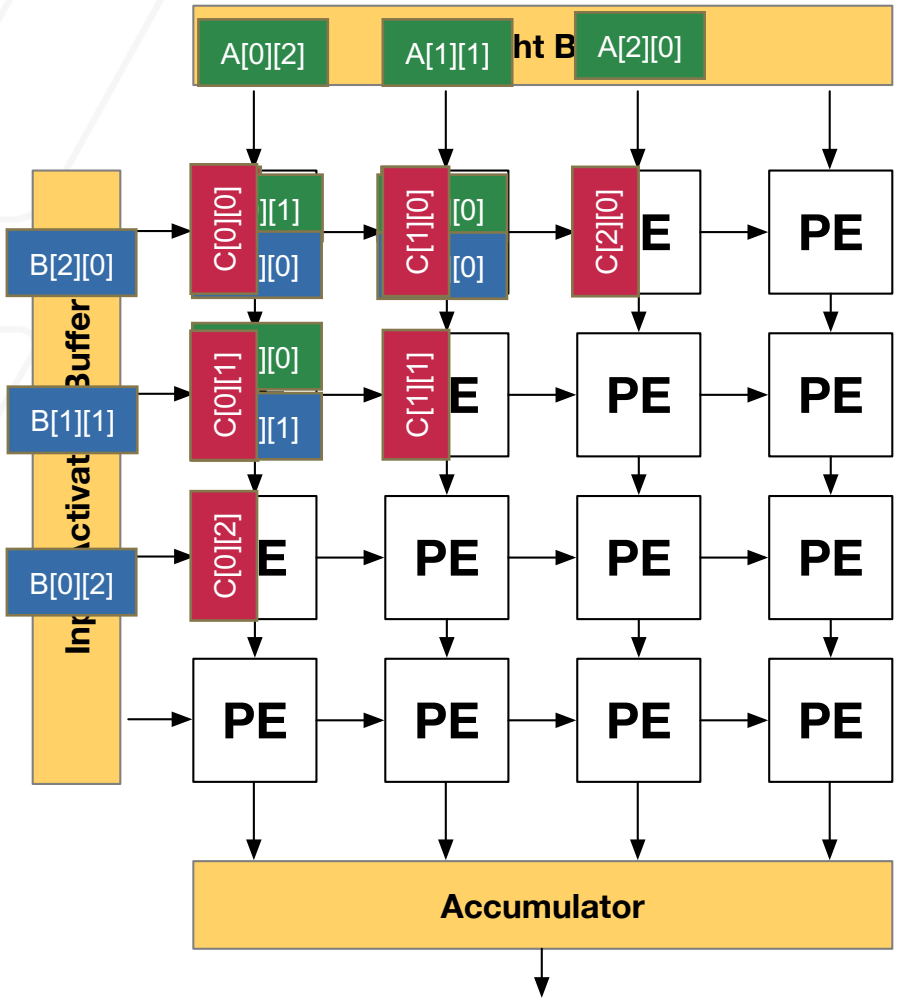

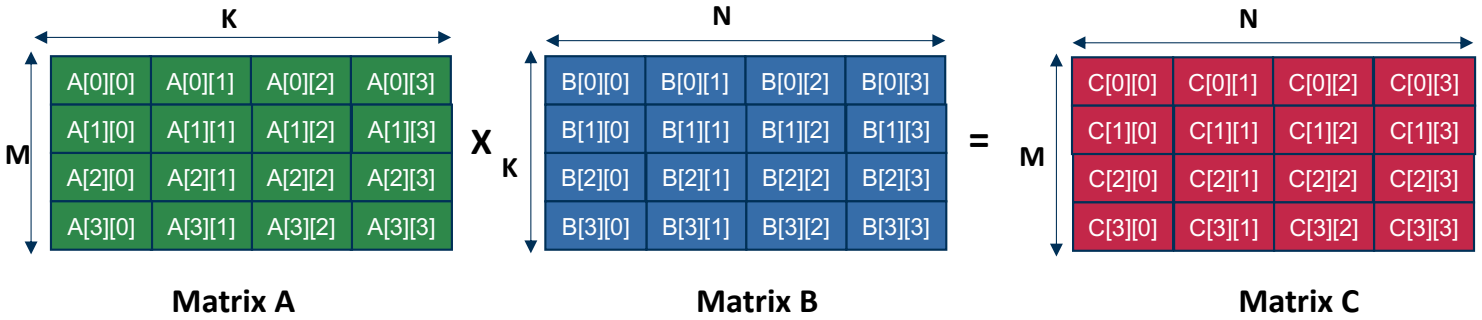


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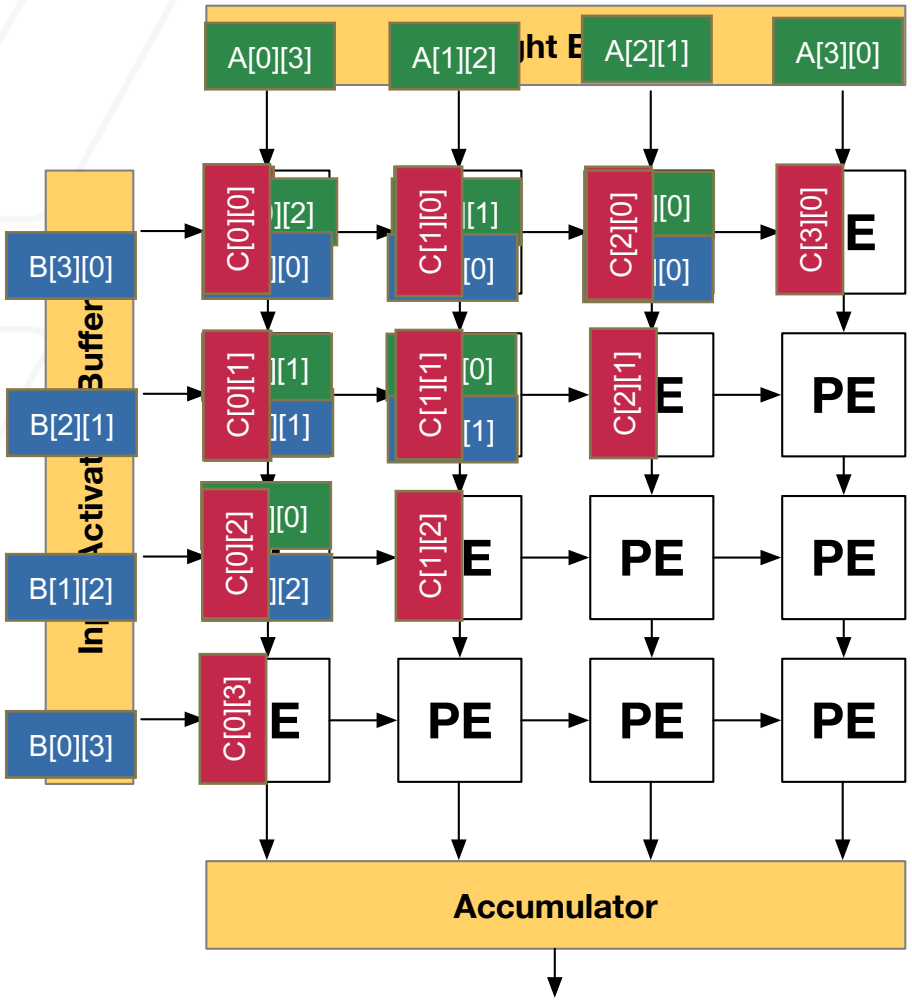

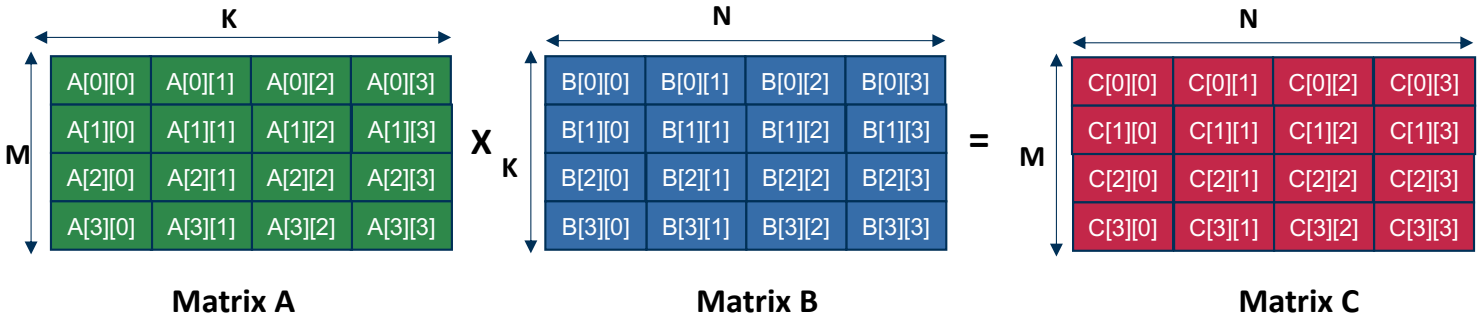


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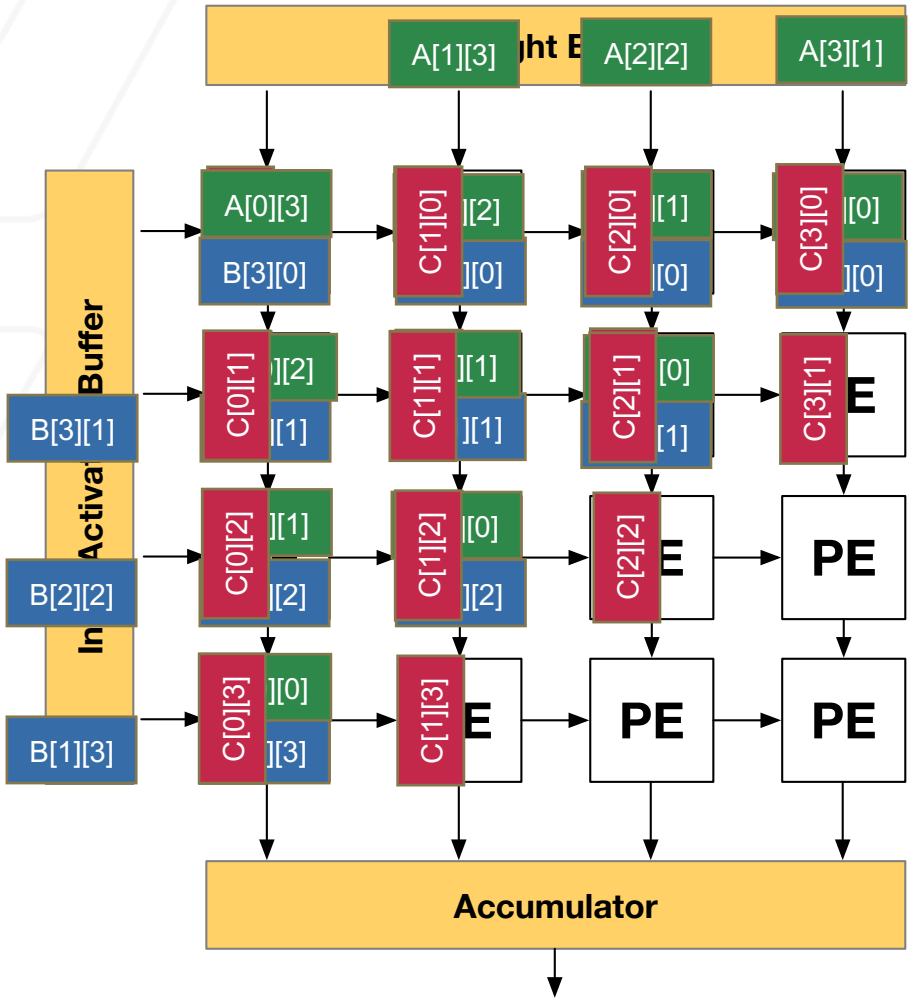

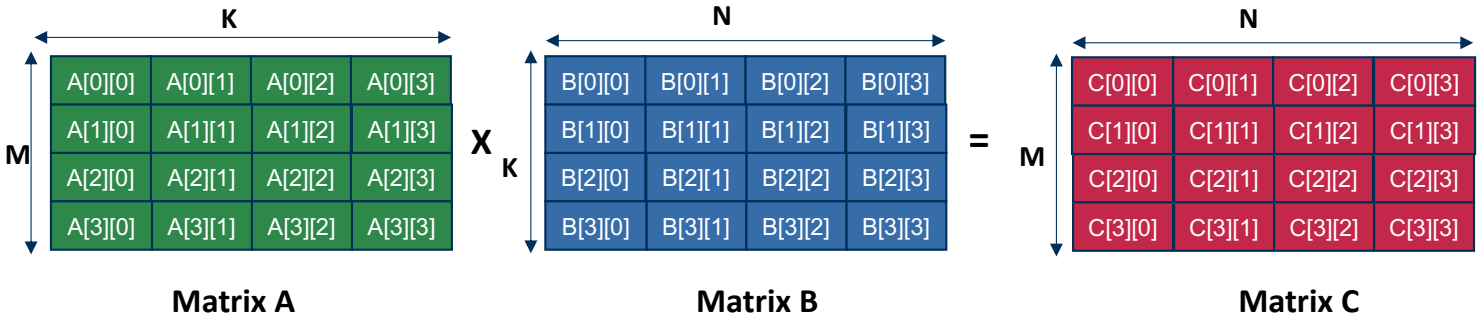


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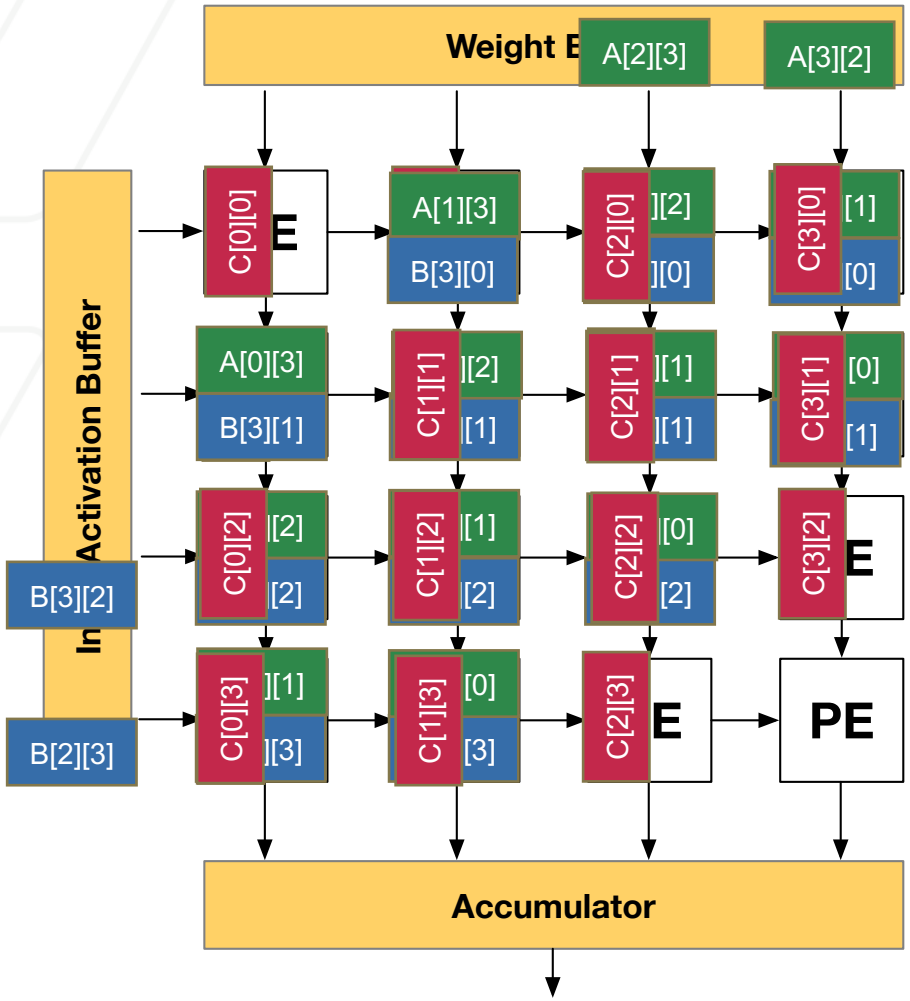

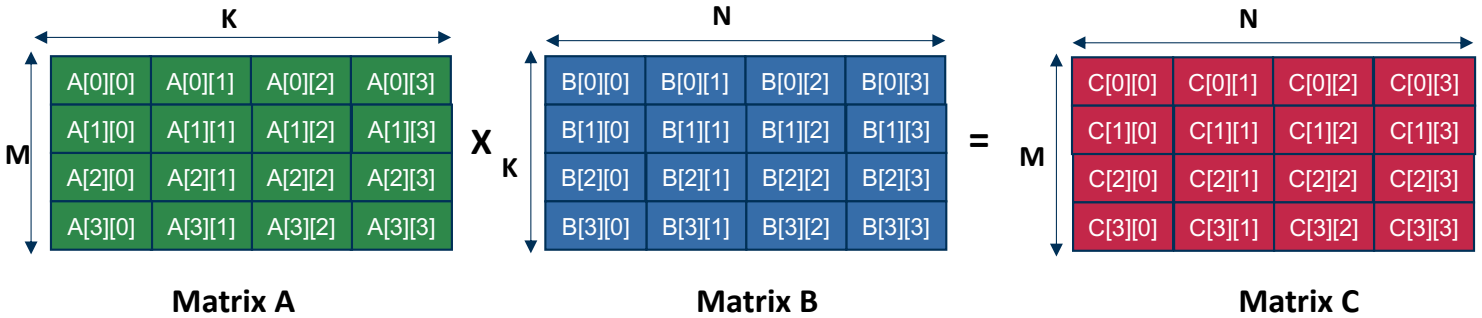


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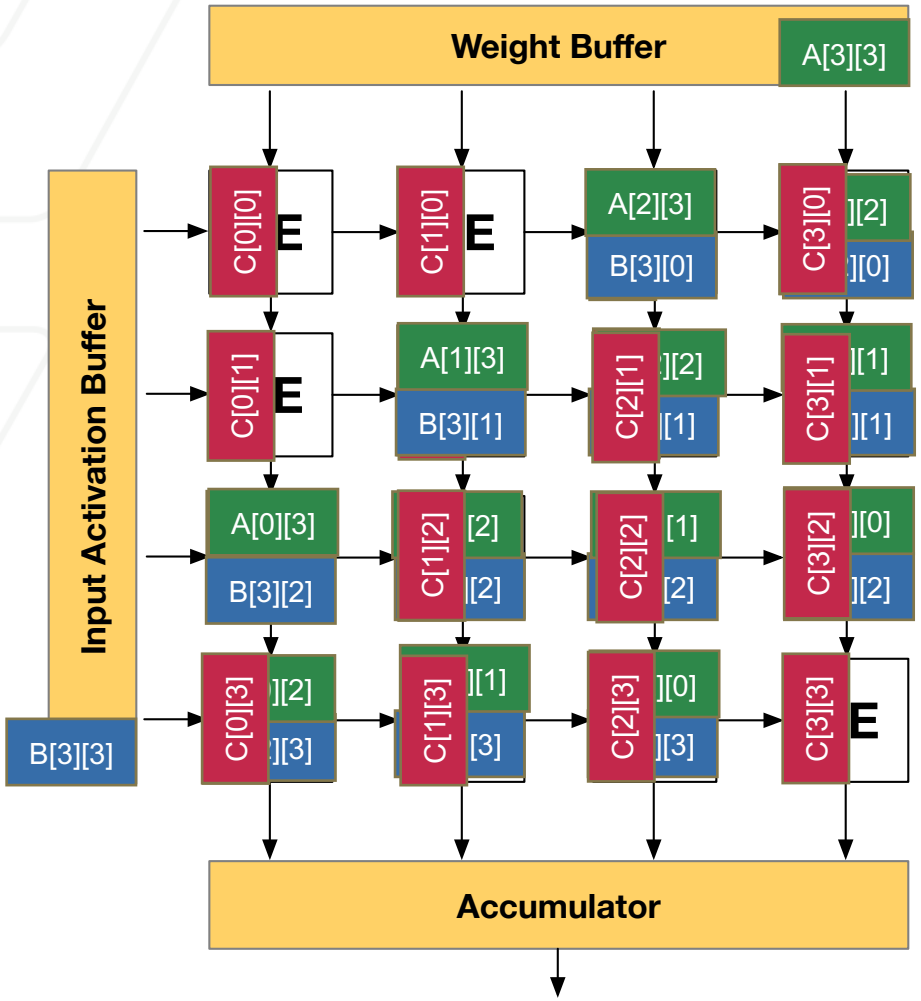

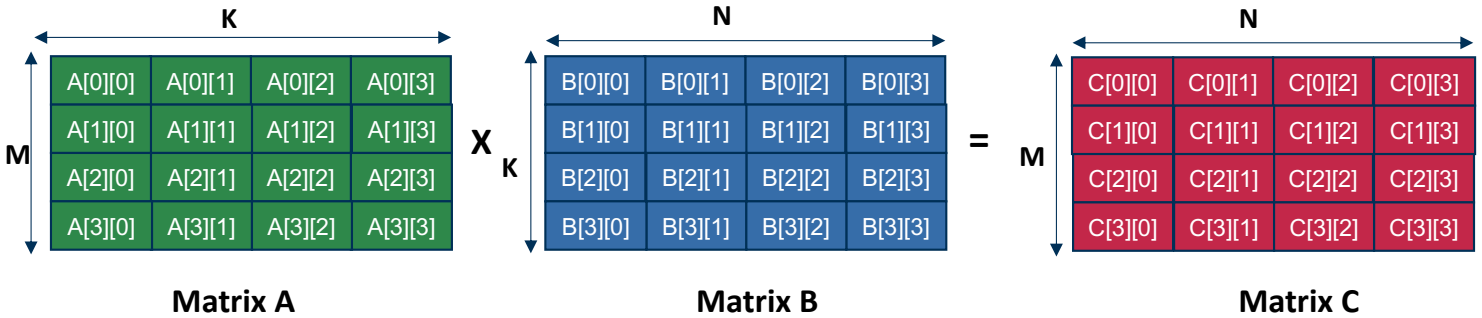


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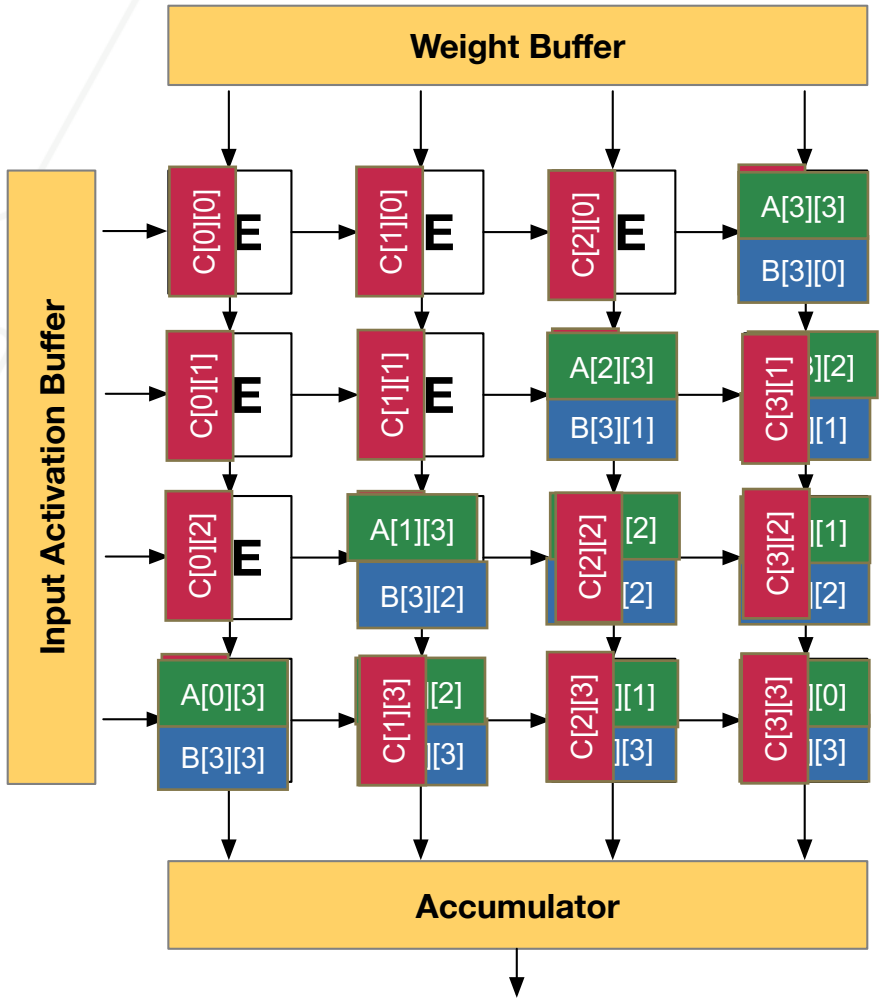

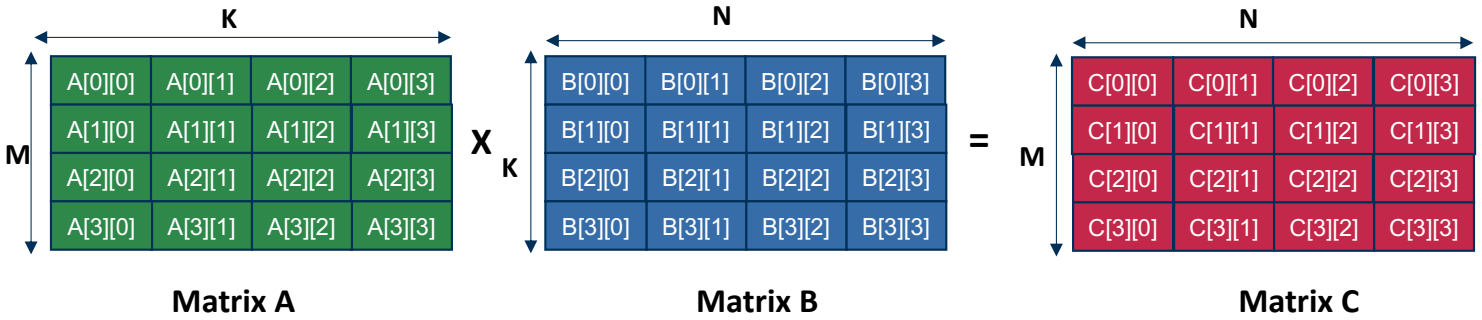


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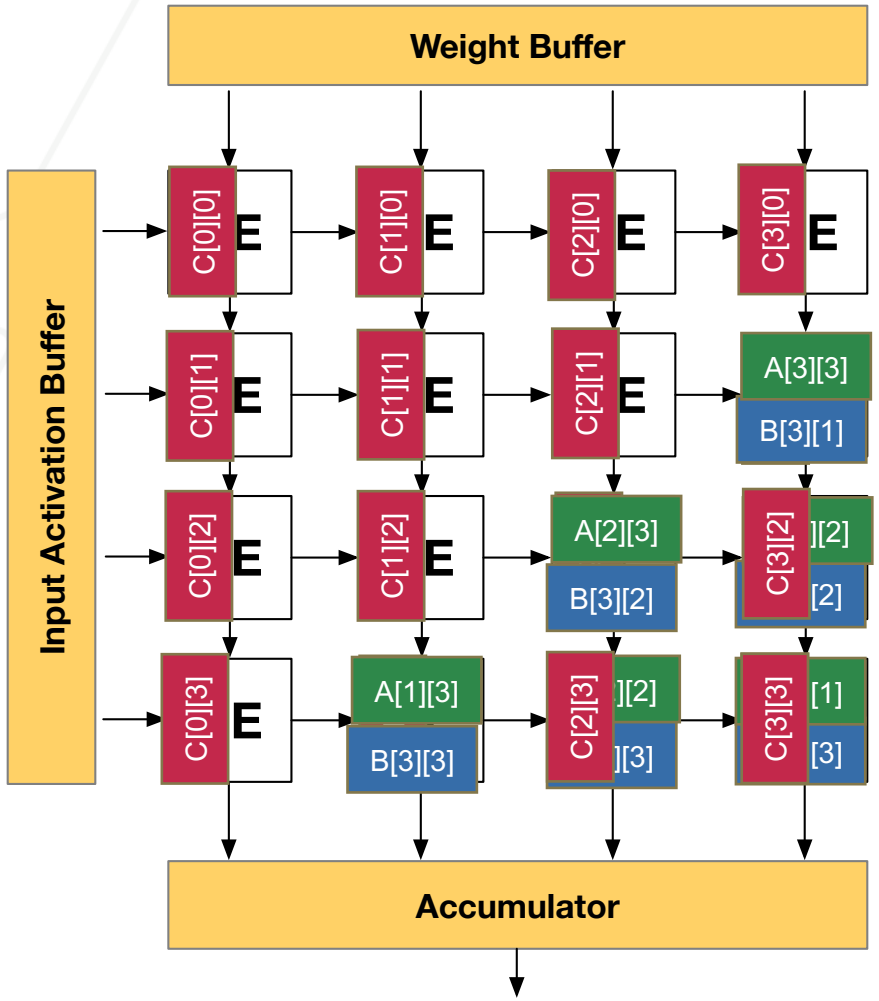

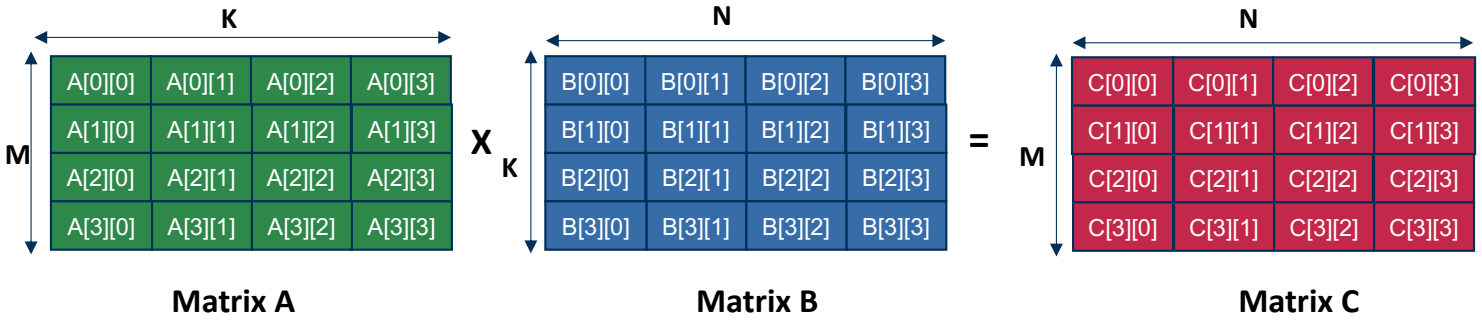


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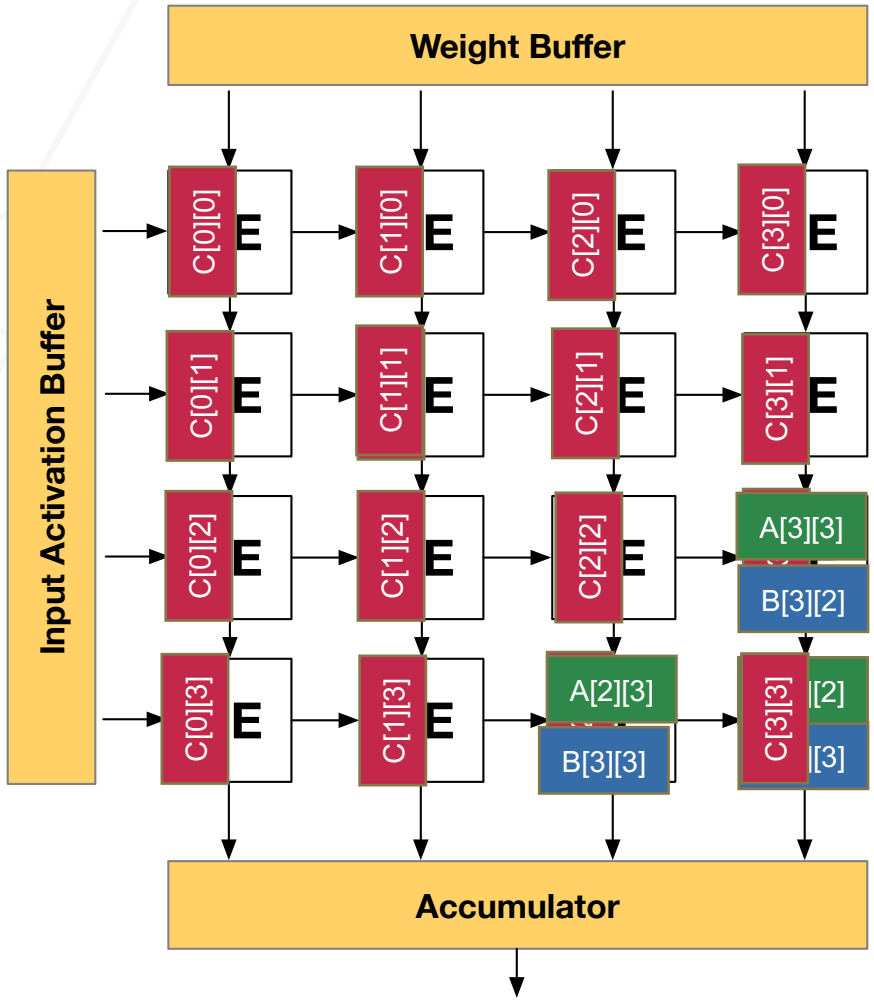

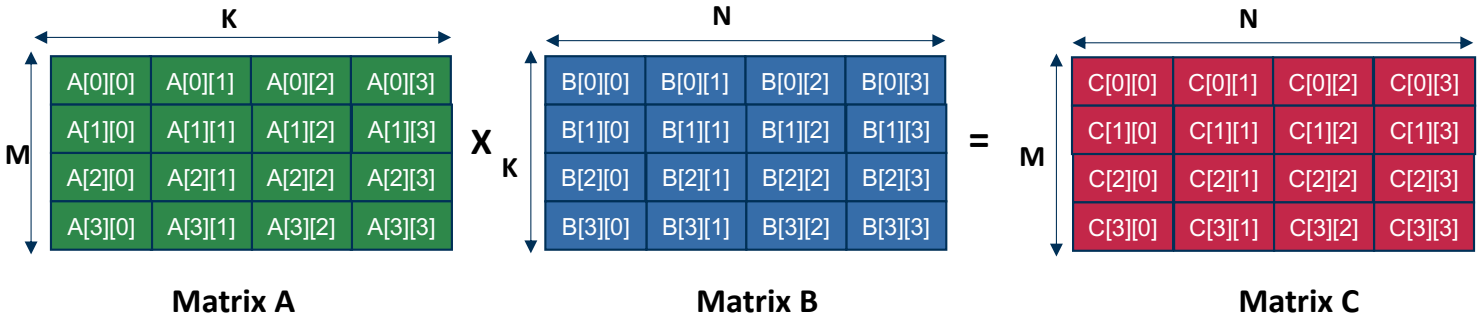


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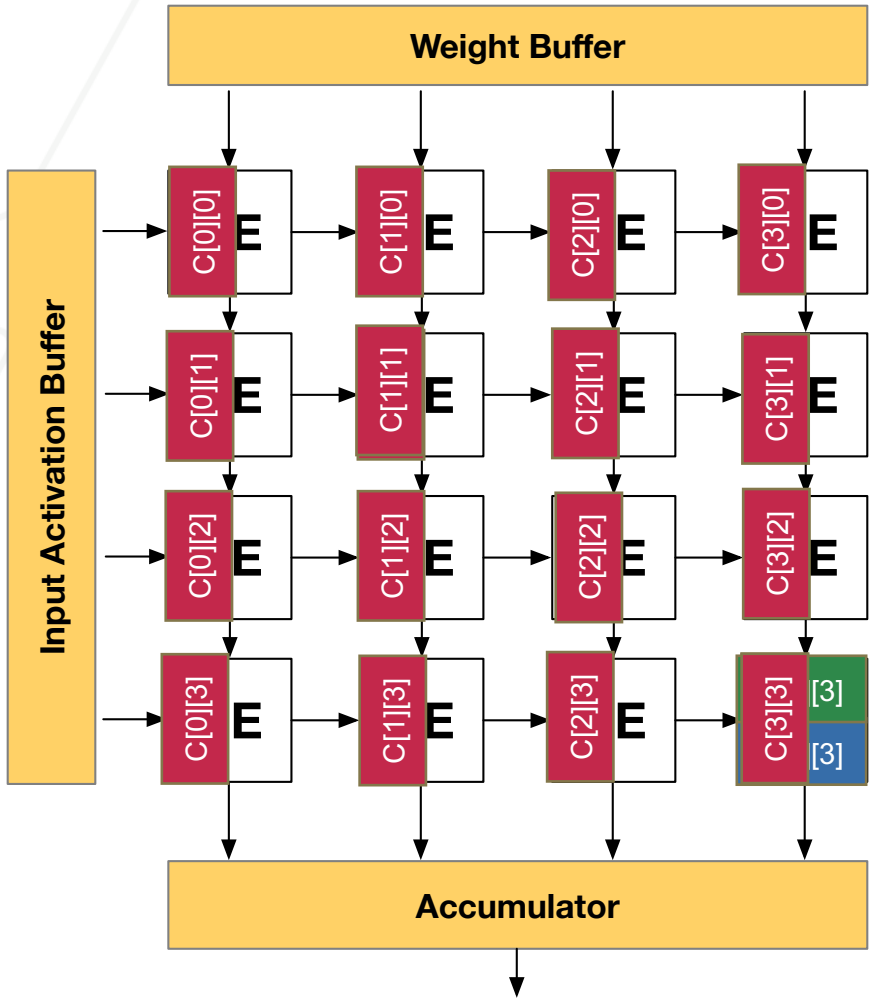

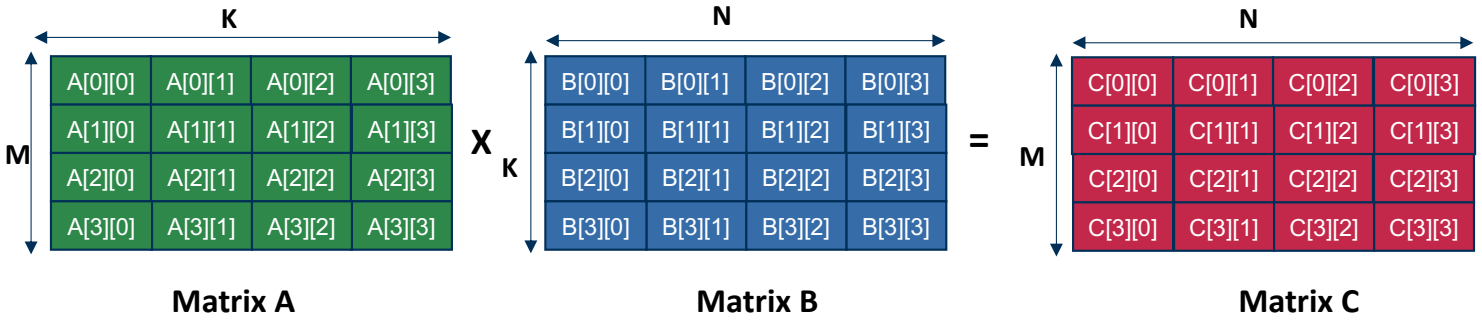


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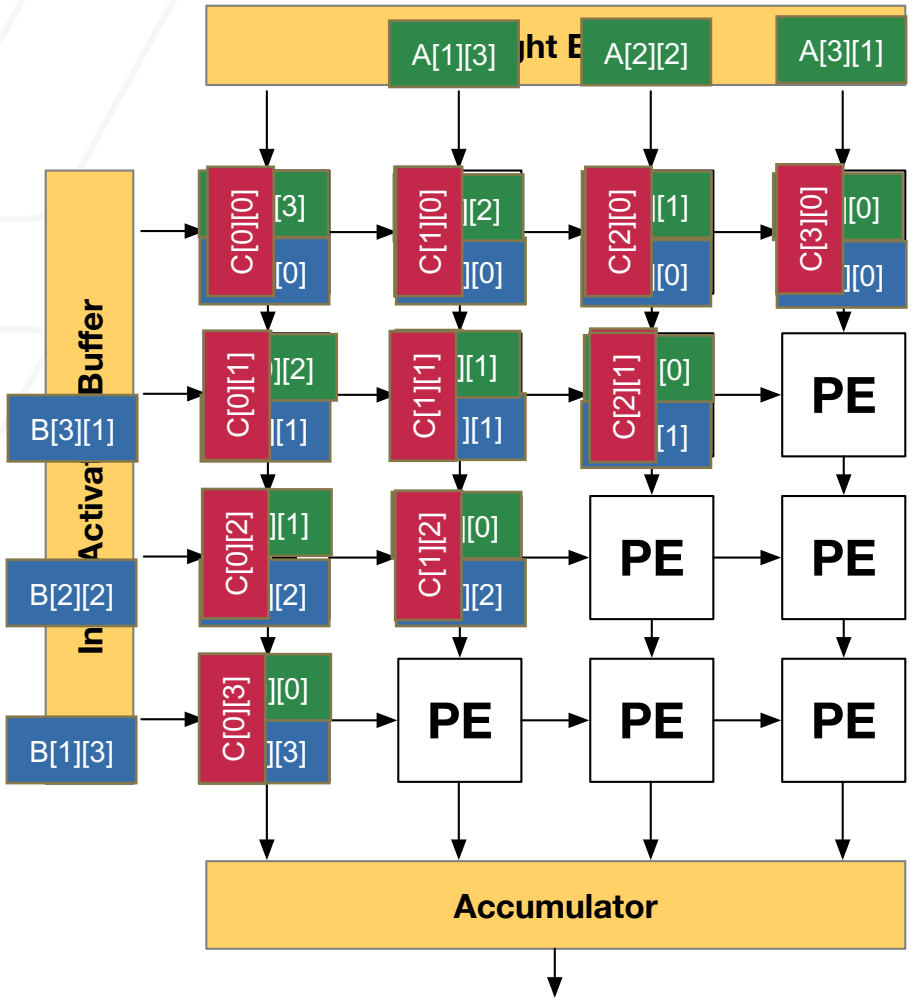
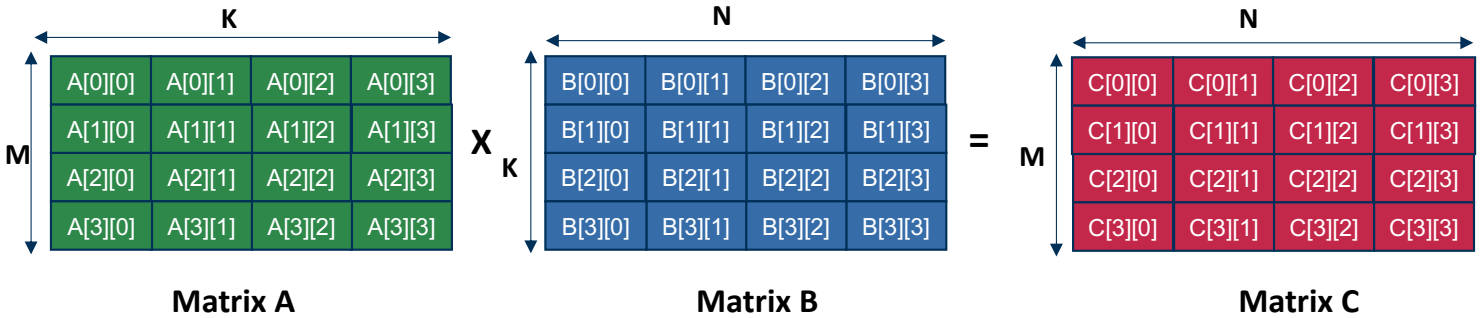
Note: need another phase to read outputs out (not shown)

Data Reuse Analysis: Qualitative





"Output-stationary" Dataflow

Tensor (A/B/C)	Reuse (Spatial/Temporal/None)
A	Spatial
B	Spatial
C	Temporal

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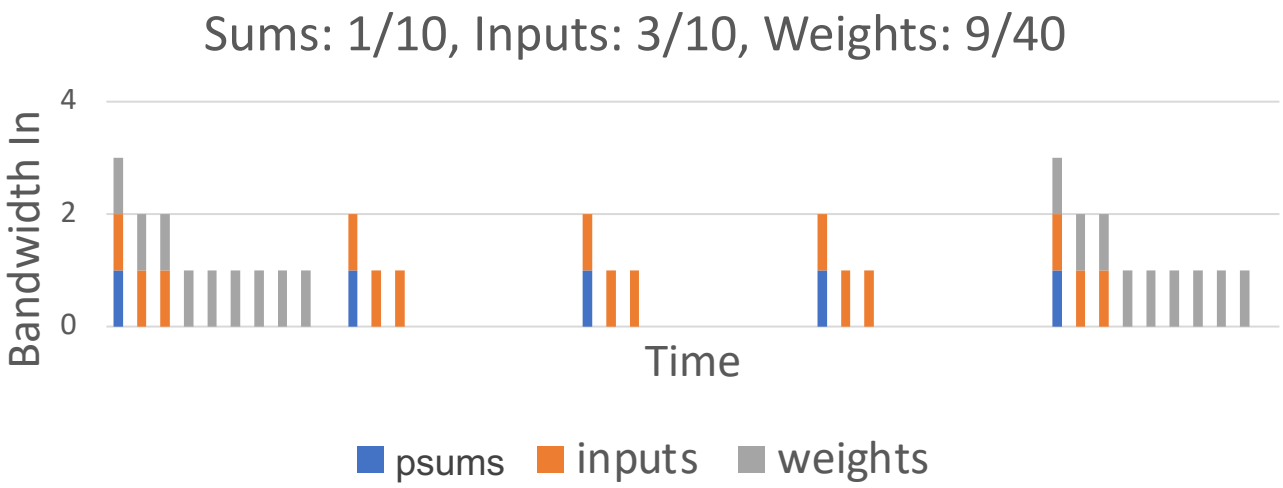
What do we mean by "stationary"?

- The datatype (and dimension) that changes most slowly



- Imprecise analogy: think of data transfers as a wave with "amplitude" and "period"
  - The stationary datatype has the **longest** period (locally held tile changes most slowly)
  - Note: like waves, also can have harmful "constructive interference" (bursts)
  - Later we will see how intermediate staging buffers reduce both bandwidth and energy
- Often corresponds to datatype that is "done with" earliest without further reloads

Note: the "stationary" name is meant to give intuition, not to be a complete specification of all the behavior of a dataflow